

Table 1 Total Variance

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	15.738	37.472	37.472	15.738	37.472	37.472	11.263	26.817	26.817
2	5.258	12.518	49.99	5.258	12.518	49.99	4.682	11.149	37.965
3	2.387	5.684	55.674	2.387	5.684	55.674	3.508	8.353	46.319
4	1.857	4.421	60.094	1.857	4.421	60.094	2.919	6.949	53.268
5	1.637	3.897	63.991	1.637	3.897	63.991	2.12	5.048	58.316
6	1.37	3.262	67.253	1.37	3.262	67.253	2.056	4.894	63.211
7	1.301	3.097	70.351	1.301	3.097	70.351	2.016	4.8	68.011
8	1.12	2.667	73.018	1.12	2.667	73.018	1.691	4.027	72.037
9	1.099	2.616	75.633	1.099	2.616	75.633	1.51	3.596	75.633
10	0.979	2.332	77.965						
11	0.892	2.124	80.089						
12	0.754	1.795	81.884						
13	0.703	1.673	83.557						
14	0.655	1.559	85.116						
15	0.633	1.506	86.622						
16	0.554	1.32	87.941						
17	0.526	1.253	89.194						
18	0.497	1.183	90.377						
19	0.47	1.119	91.496						
20	0.438	1.042	92.539						
21	0.399	0.951	93.49						
22	0.36	0.858	94.348						
23	0.314	0.748	95.096						
24	0.284	0.676	95.772						
25	0.247	0.589	96.361						
26	0.219	0.522	96.883						
27	0.199	0.475	97.358						
28	0.182	0.433	97.792						
29	0.177	0.421	98.213						
30	0.146	0.348	98.561						
31	0.122	0.291	98.851						
32	0.101	0.24	99.092						
33	0.086	0.205	99.296						
34	0.066	0.157	99.454						
35	0.055	0.131	99.585						
36	0.054	0.128	99.713						
37	0.041	0.097	99.81						
38	0.029	0.07	99.88						
39	0.022	0.053	99.933						
40	0.016	0.038	99.971						
41	0.009	0.021	99.992						
42	0.003	0.008	100						

Fig. 1 Scree Plot

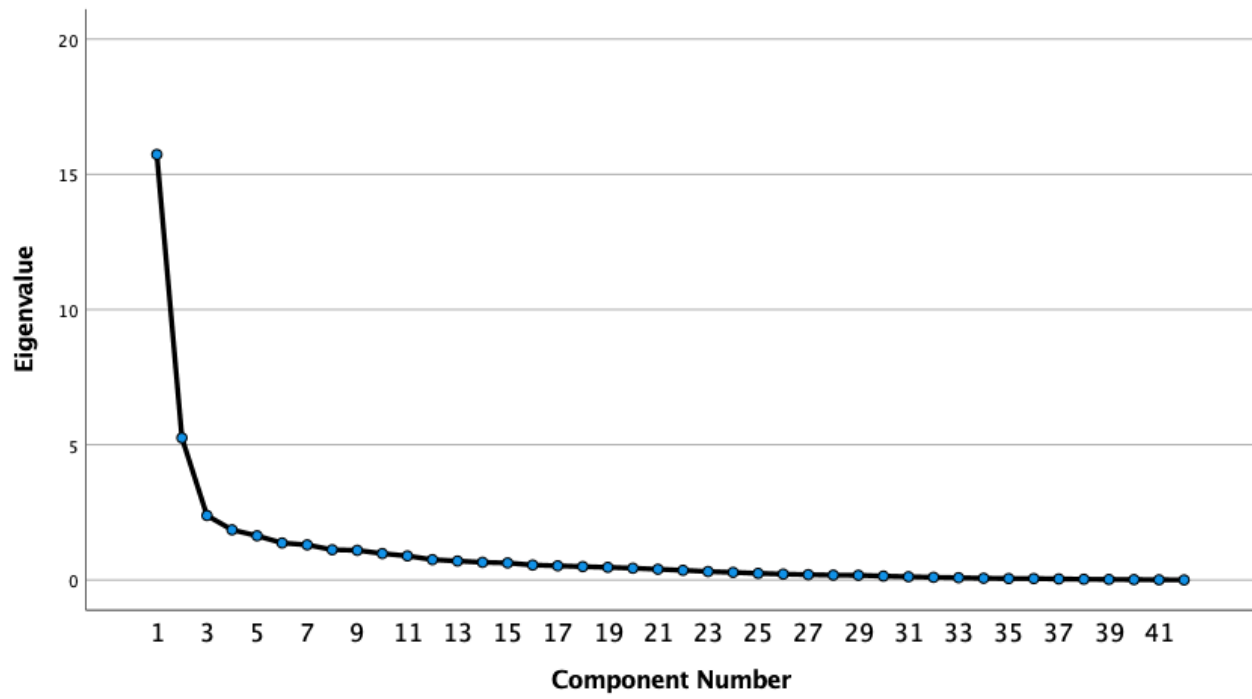


Table 2 Rotated Component Matrix

	Component								
	1	2	3	4	5	6	7	8	9
Gene38	0.86								
Gene16	0.835								
Gene42	0.796								
Gene01	0.783								
Gene19	0.738								
Gene20	0.737								
Gene41	0.702								
Gene31	0.694								
Gene30	0.684								
Gene37	0.684								
Gene03	0.673	0.408							
Gene13	0.671								
Gene40	0.641			0.489					
Gene32	0.629		0.598						
Gene29	0.61	0.457							
Gene25	0.609						0.443		
Gene15	0.604	0.455							
Gene26	0.572	0.467							
Gene35	0.539		0.525						
Gene04	0.478			0.412					
Gene14	0.458			0.435					
Gene12	0.408								
Gene17		0.788							
Gene39		0.695							
Gene05		0.671							
Gene08		0.623							
Gene27		0.569							
Gene11		0.533						0.406	
Gene33			0.813						
Gene36			0.683						
Gene34		0.501	0.663						
Gene21	0.432			0.710					
Gene24	0.481			0.583					
Gene09	0.408			0.482					
Gene18					0.839				
Gene02					0.804				
Gene10						0.807			
Gene28							0.738		
Gene23			-0.438				0.58		
Gene07								0.699	
Gene06	0.538							-0.566	
Gene22									0.789

Fig. 2 Component Plot in Rotated Space

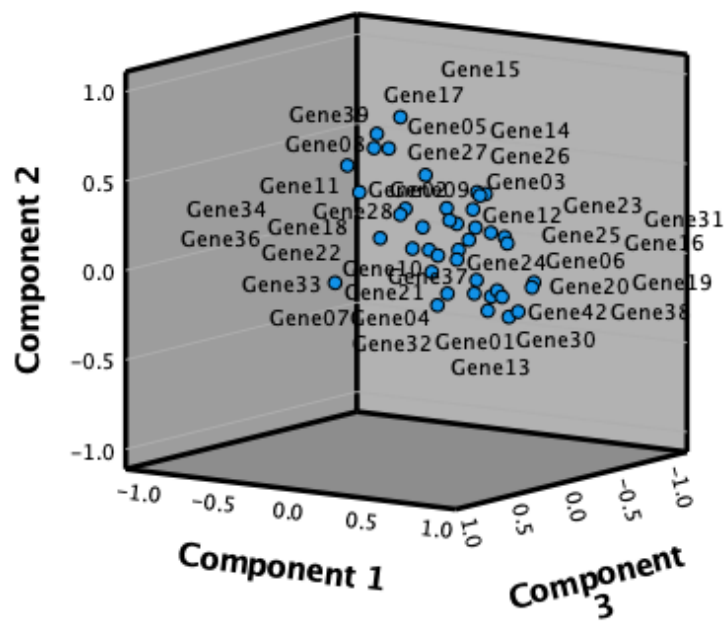


Fig. 3 Component Plot in 2D Space

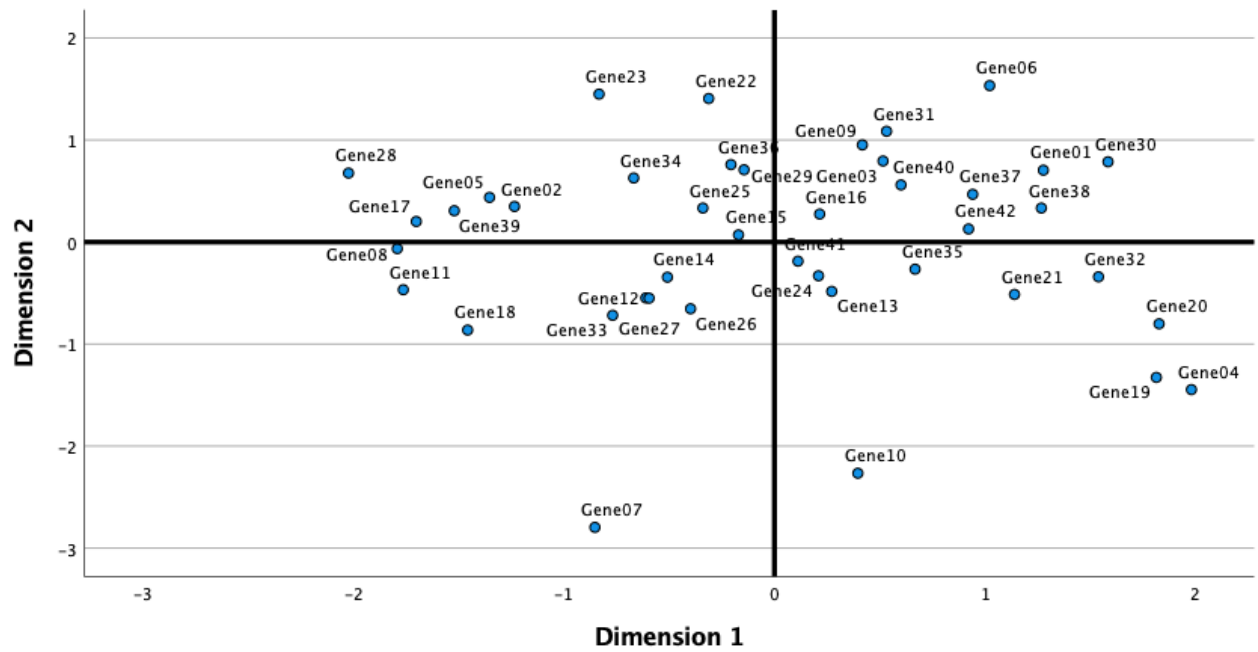


Fig. 4 Euclidean Distance Model

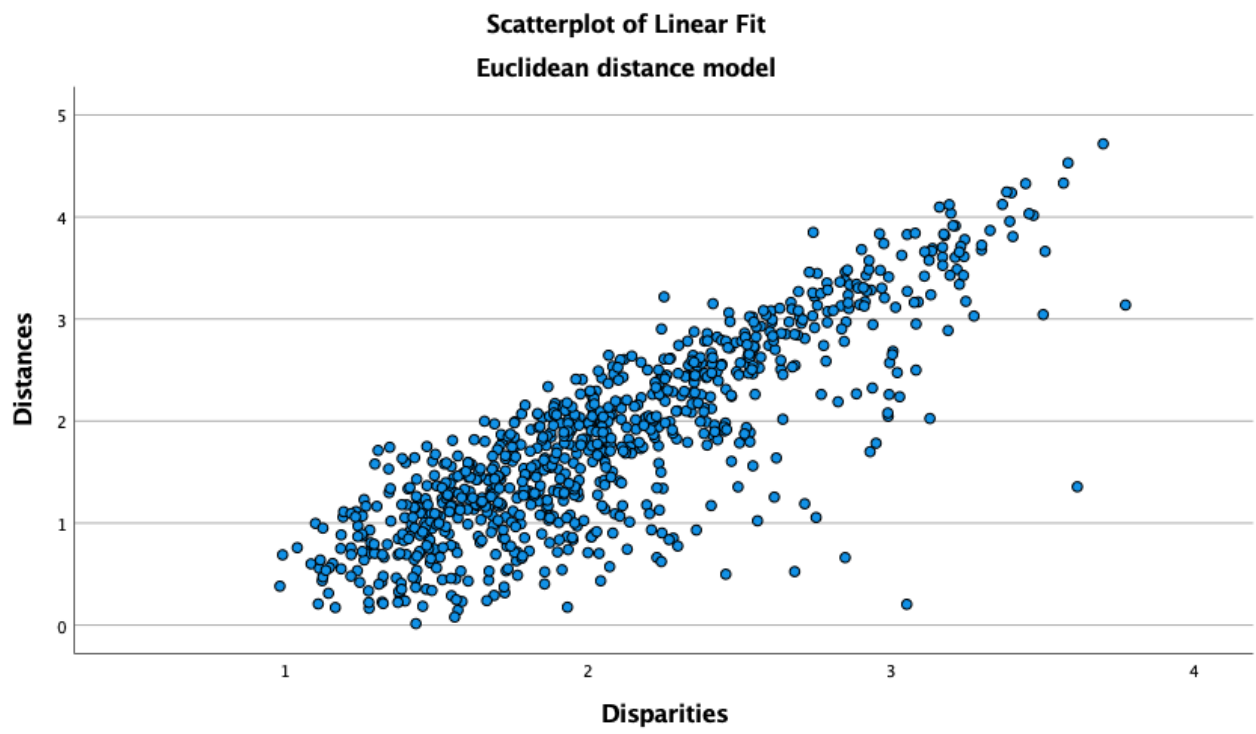


Fig. 5 Dendrogram

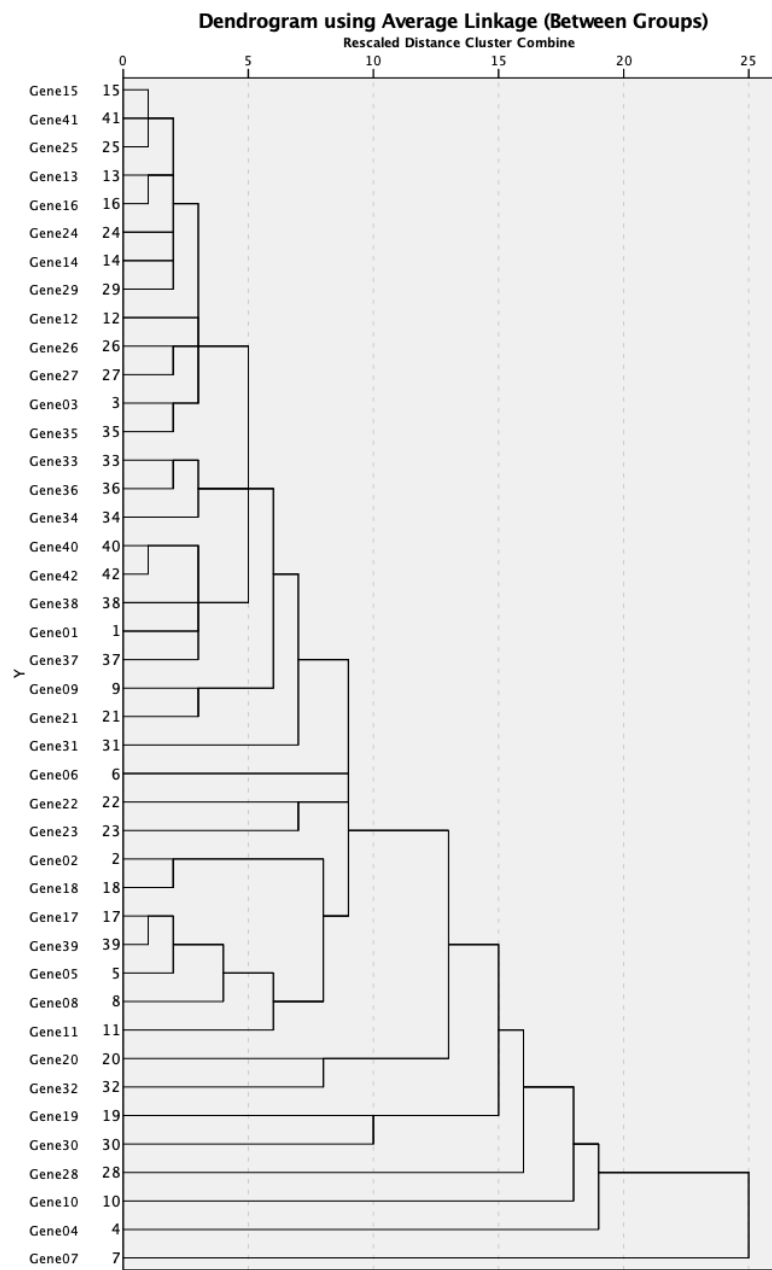
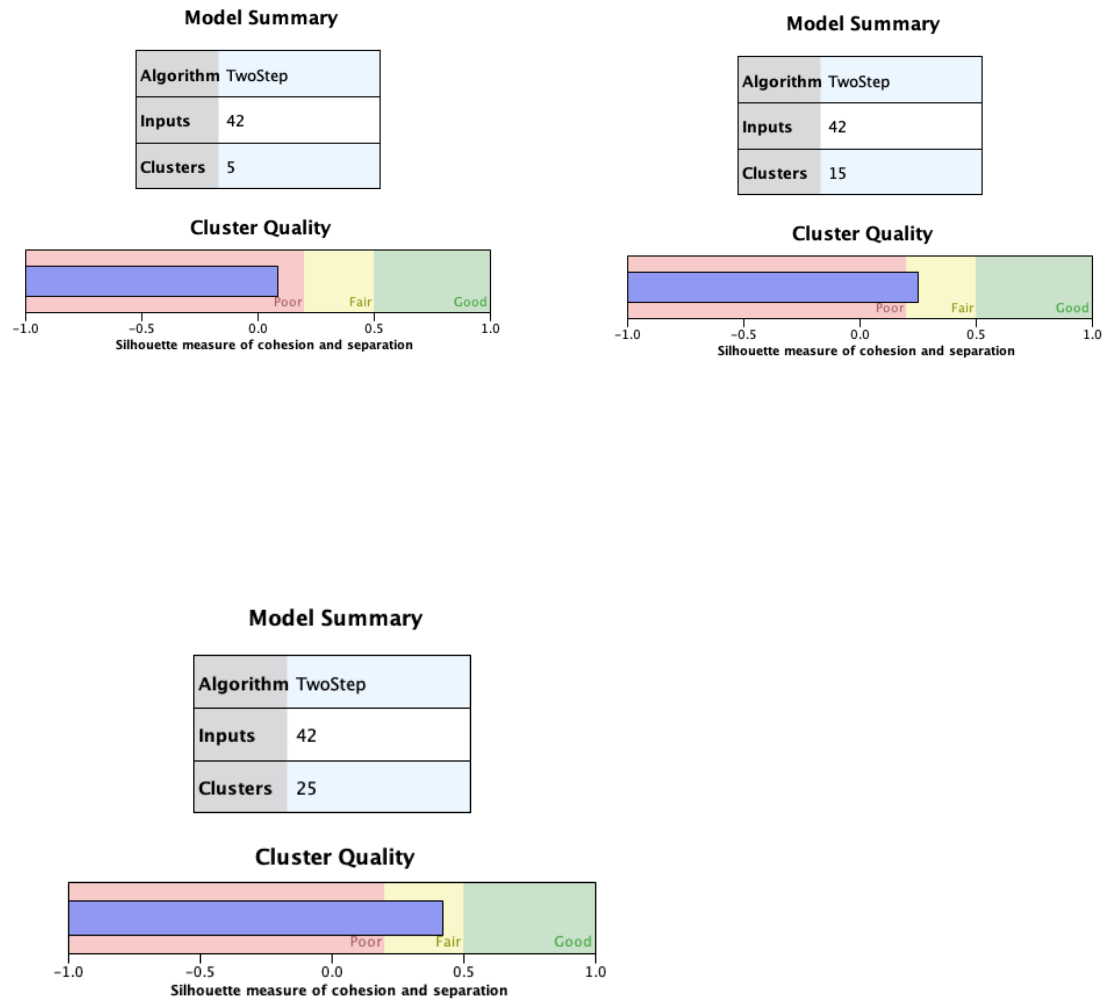


Fig. 6 Two-Step Cluster



Footnote: Analysis on clusters with 5, 15 and 25 clusters.

Table 3 Reliability Analysis

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Gene01	-7.3860044	4126.173	0.708	0.940
Gene02	-6.5492509	4252.158	0.493	0.941
Gene03	-7.4383266	4160.332	0.789	0.939
Gene04	-7.1108386	4026.378	0.607	0.941
Gene05	-5.5745065	4308.530	0.419	0.942
Gene06	-6.6619547	4196.066	0.479	0.941
Gene07	-6.1947866	4195.693	0.308	0.945
Gene08	-5.9155379	4393.468	0.059	0.944
Gene09	-7.471386	4181.088	0.682	0.940
Gene10	-7.628132	4150.654	0.459	0.942
Gene11	-6.8468688	4355.049	0.144	0.944
Gene12	-6.3367099	4241.252	0.639	0.941
Gene13	-6.3815383	4241.289	0.648	0.941
Gene14	-6.241345	4213.879	0.714	0.940
Gene15	-7.0245703	4195.455	0.801	0.940
Gene16	-6.6735475	4197.224	0.820	0.940
Gene17	-5.9046929	4380.739	0.106	0.943
Gene18	-6.3949393	4206.810	0.462	0.942
Gene19	-7.3558288	4143.683	0.466	0.942
Gene20	-7.8938951	4107.153	0.589	0.941
Gene21	-7.9157553	4175.811	0.623	0.940
Gene22	-7.3683303	4293.527	0.342	0.942
Gene23	-7.121017	4364.467	0.128	0.944
Gene24	-7.0223159	4206.340	0.716	0.940
Gene25	-6.3664547	4174.350	0.763	0.940
Gene26	-6.7093854	4164.997	0.748	0.940
Gene27	-6.8723933	4188.807	0.683	0.940
Gene28	-6.3206343	4332.057	0.158	0.944

Gene29	-6.4278116	4170.317	0.742	0.940
Gene30	-7.5165876	4218.875	0.428	0.942
Gene31	-6.5036477	4198.804	0.555	0.941
Gene32	-7.5783456	4135.207	0.603	0.940
Gene33	-6.3686997	4277.185	0.443	0.942
Gene34	-7.0045	4274.025	0.461	0.941
Gene35	-7.3000365	4175.220	0.698	0.940
Gene36	-7.0044712	4214.429	0.609	0.941
Gene37	-7.8246483	4166.296	0.711	0.940
Gene38	-7.8439594	4138.818	0.707	0.940
Gene39	-6.0069334	4353.155	0.193	0.943
Gene40	-6.8868916	4210.227	0.655	0.940
Gene41	-6.649778	4156.879	0.838	0.939
Gene42	-6.9496264	4194.401	0.650	0.940

Table 4 Rotated Component Matrix: Identified Genes

Biomarker	Gene	Component								
		1	2	3	4	5	6	7	8	9
Gene38	TGFB3	0.860								
Gene16	CDKN1B	0.835								
Gene42	VEGFC	0.796								
Gene01	BAD	0.783								
Gene19	E2F3	0.738								
Gene20	EGF	0.737								
Gene41	TSC2	0.702								
Gene31	PDGFRB	0.694								
Gene30	PDGFC	0.684								
Gene37	TGFB1	0.684								
Gene03	BCL2	0.673	0.408							
Gene13	CDK4	0.671								
Gene40	TLR2	0.641			0.489					
Gene32	P1K3CA	0.629		0.598						
Gene29	MMP2	0.610	0.457							
Gene25	IRS2	0.609						0.443		
Gene15	CDKN1A	0.604	0.455							
Gene26	KRAS	0.572	0.467							
Gene35	SMAD4	0.539		0.525						
Gene04	BCL2L1	0.478			0.412					
Gene14	CDK6	0.458			0.435					
Gene12	CDK2	0.408								
Gene17	CREB1		0.788							
Gene39	TGFB2		0.695							
Gene05	BID		0.671							
Gene08	CCND2		0.623							
Gene27	MAPK3		0.569							
Gene11	CDH1		0.533						0.406	
Gene33	PIK3R3			0.813						
Gene36	STAT1			0.683						
Gene34	SHC1		0.501	0.663						
Gene21	EGFR	0.432			0.710					
Gene24	HRAS	0.481			0.583					
Gene09	CCND3	0.408			0.482					
Gene18	CTNNB1					0.839				
Gene02	BAX					0.804				
Gene10	CCNE1						0.807			
Gene28	MDM2							0.738		
Gene23	FN1			-0.438				0.580		
Gene07	CCND1								0.699	
Gene06	CCNA1	0.538							-0.566	
Gene22	FGFR1									0.789

Table 5 Summary of Gene Functions across Components

Group	Main Function or Relevant Molecular Phenomena	C1	C2	C3	C4	C5	C6	C7	C8	C9
A	Regulation of cell cycle	x	x		x		x		x	
B	Cell growth, proliferation, differentiation or embryogene	x	x	x	x					x
C	Regulation of apoptosis and cell death	x	x			x				
D	Transcription factors		x	x						
E	Cell adhesion, motility and/or shape		x			x		x		
F	Growth factors and their receptors	x								
G	Cellular signaling, signal transduction		x	x						
H	Tumor suppressor genes									
I	Extracellular metalloproteinase									
J	Immune system regulation									
K	Ubiquitin-protein ligase							x		
	Total	4	6	3	2	2	1	2	1	1

Table 6 Identification of Significant Genes (Component 1)

Function by Group	Main Function or Relevant molecular Phenomena	C1	C1 (%)
A	Regulation of cell cycle	5	22.7%
B	Cell growth, proliferation, differentiation or embryogenesis, wound healing	5	22.7%
C	Regulation of apoptosis and cell death	3	13.6%
F	Growth factors and their receptors	3	13.6%
D	Transcription factors	2	9.1%
G	Cellular signaling, signal transduction	1	4.5%
H	Tumor suppressor genes	1	4.5%
I	Extracellular metalloproteinase	1	4.5%
J	Immune system regulation	1	4.5%

Table 7 Identification of Significant Genes (Component 2)

Function by Group	Main Function or Relevant molecular Phenomena	C2	C2 (%)
A	Regulation of cell cycle	1	16.7%
B	Cell growth, proliferation, differentiation or embryogenesis, wound healing	1	16.7%
C	Regulation of apoptosis and cell death	1	16.7%
D	Transcription factors	1	16.7%
E	Cell adhesion, motility and/or shape	1	16.7%
G	Cellular signalling, signal transduction	1	16.7%

Table 8 Identification of Significant Genes (Component 3)

Function by Group	Main Function or Relevant molecular Phenomena	C3	C3 (%)
B	Cell growth, proliferation, differentiation or embryogenesis, wound healing	1	33.3%
D	Transcription factors	1	33.3%
G	Cellular signalling, signal transduction	1	33.3%

Table 9 Identification of Significant Genes (Component 4)

Function by Group	Main Function or Relevant molecular Phenomena	C4	C4 (%)
B	Cell growth, proliferation, differentiation or embryogenesis, wound healing	2	66.7%
A	Regulation of cell cycle	1	33.3%

Table 10 Identification of Significant Genes (Component 5)

Function by Group	Main Function or Relevant molecular Phenomena	C5	C5 (%)
C	Regulation of apoptosis and cell death	1	50.0%
E	Cell adhesion, motility and/or shape	1	50.0%

Table 11 Identification of Significant Genes (Component 6)

Function by Group	Main Function or Relevant molecular Phenomena	C6	C6 (%)
A	Regulation of cell cycle	1	100.0%

Table 12 Identification of Significant Genes (Component 7)

Function by Group	Main Function or Relevant molecular Phenomena	C7	C7 (%)
E	Cell adhesion, motility and/or shape	1	50.0%
K	Ubiquitin-protein ligase	1	50.0%

Table 13 Identification of Significant Genes (Component 8)

Function by Group	Main Function or Relevant molecular Phenomena	C8	C8 (%)
A	Regulation of cell cycle	2	100.0%

Table 14 Identification of Significant Genes (Component 9)

Function by Group	Main Function or Relevant molecular Phenomena	C9	C9 (%)
B	Cell growth, proliferation, differentiation or embryogenesis, wound healing	1	100.0%

Table 15 Multifunctional Biomarkers Identified: Component 1

	C1* and C2**	C1* and C3**	C1* and C4**	C1* and C7**
Gene Function	C, I, A, B	B, D	J, C, A	G
Biomarker	3, 29, 15, 26	32, 35	40, 4, 14	25

Footnote:* Primary function and ** Secondary function

Table 16 Multifunctional Biomarkers Identified: Components 2 to 8

	C2* and C8**	C3* and C2**	C4* and C1**	C7* and C3**	C8* and C1**
Gene Function	E	G	B, A	E	A
Biomarker	11	34	B: 21, 24 A: 9	23	6

Footnote:* Primary component and ** Secondary component

Table 17 Biomarkers Selected for PCA

	Identified Genes	Function by Group	Coefficient	Component
Gene38	TGFB3	B	0.860	1
Gene16	CDKN1B	A	0.835	1
Gene42	VEGFC	F	0.796	1
Gene01	BAD	C	0.783	1
Gene19	E2F3	D	0.738	1
Gene20	EGF	B	0.737	1
Gene41	TSC2	H	0.702	1
Gene17	CREB1	D	0.788	2
Gene33	PIK3R3	B	0.813	3
Gene21	EGFR	B	0.710	4

Table 18 Total Variance

Comp onent	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Varian	Cumul ative	Total	% of Varian	Cumul ative	Total	% of Varian	Cumul ative
1	5.168	51.67	51.67	5.168	51.67	51.67	5.162	51.62	51.62
2	1.255	12.55	64.22	1.255	12.55	64.22	1.261	12.60	64.22
3	0.969	9.685	73.91						
4	0.68	6.798	80.71						
5	0.471	4.714	85.42						
6	0.396	3.959	89.38						
7	0.384	3.839	93.22						
8	0.288	2.881	96.10						
9	0.225	2.249	98.35						
10	0.165	1.645	100						

Table 19 Rotation Component Matrix

Biomarker	Identified Gene	Function by Group	Component	
			1	2
Gene38	TGFB3	B	0.880	
Gene42	VEGFC	F	0.852	
Gene16	CDKN1B	A	0.845	
Gene01	BAD	C	0.829	
Gene41	TSC2	H	0.815	
Gene20	EGF	B	0.792	
Gene19	E2F3	D	0.670	-0.339
Gene21	EGFR	B	0.643	
Gene17	CREB1	D		0.705
Gene33	PIK3R3	B		0.661

Fig. 7 Scree Plot

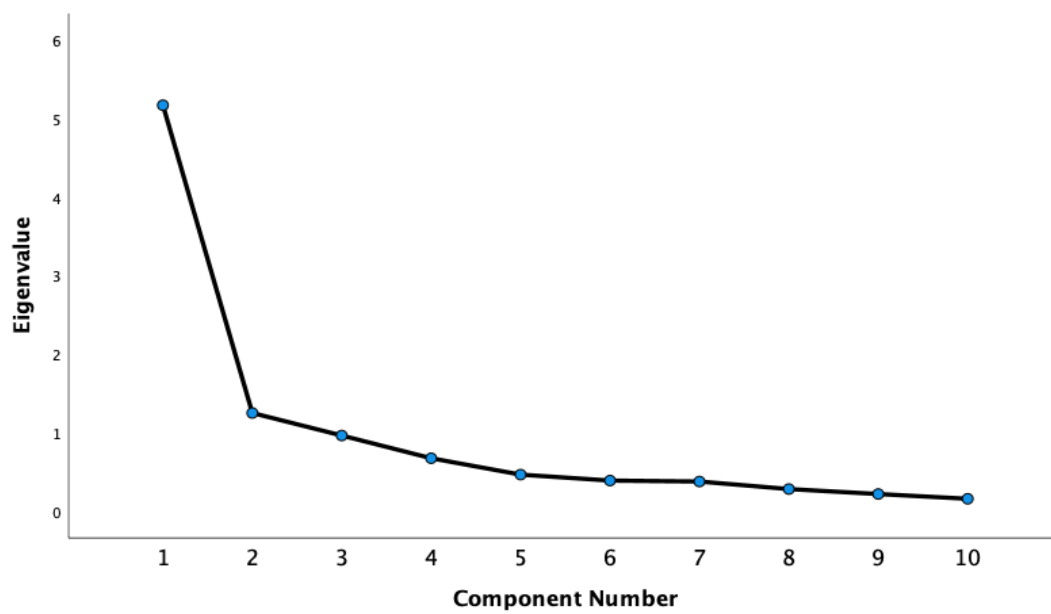


Table 20 Bootstrap (BCa) on 42 Biomarkers

[illegible]

Footnote: Pearson correlation, n = 27500. Cells that are highlighted red are instances where correlation between biomarkers are significantly high (exceed 0.60).

Table 21 Bootstrap (Percentile) on 42 Biomarkers

[illegible]

Footnote: Pearson correlation, $n = 500$

Table 21b Included and Excluded Genes with Gene Functionalities

Gene	Average DDct	Gene Identified	Function of Gene	Reason for Inclusion/ Exclusion
Gene05	-1.4632224	BID	C	Limitation: Not in top 10 expression ratio Additional benefits: Most negative average DDct
Gene17	-1.133036	CREB1	D	Limitation: Function is different and has a strong effect Additional benefits: In top 10 expression ratio
Gene18	-0.6427895	CTNNB1	E	Additional benefit: Function different but does not have strongest effect Not in top 10 expression ratio
Gene28	-0.7170946	MDM2	K	Limitation: Not in top 10 expression ratio/ function different but does not have strongest effect Additional benefits: MDM2 is one of the most important target for cancer treatment in the p53 pathway (Wang et al. 2021; Hou et al. 2019)
Gene25	-0.6712741	IRS2	G	Limitation: Not in top 10 expression ratio Additional benefit: Function different but does not have strongest effect
Gene29	-0.6099172	MMP2	I	Additional benefit: Function different but does not have strongest effect Not in top 10 expression ratio
Gene33	-0.6690292	PIK3R3	B	Limitation: Not in top 10 expression ratio and not strongest effect Additional benefit: Function similar so between two least effective genes (IRS2 and PIK3R3, so the PIK3R3 gene was excluded)

Table 22 DDCt Selected and Excluded Genes

Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession	Gene	Protein	Accession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Table 23 Selected Genes

Gene	Average DDCtTrans	Previous Top 10
Gene05	-1.463222396	
Gene07	-0.84294231	Y
Gene08	-1.122190935	
Gene12	-0.701018938	
Gene13	-0.656190577	
Gene14	-0.796383848	
Gene17	-1.133035977	
Gene18	-0.642789546	
Gene20	0.856166215	
Gene21	0.878026455	Y
Gene25	-0.671274135	
Gene28	-0.71709458	
Gene29	-0.609917239	
Gene33	-0.669029191	
Gene37	0.786919468	Y
Gene38	0.806230549	Y
Gene39	-1.030795501	Y

Table 24 Excluded Genes

Gene	Average DDctTrans	Previous Top 10
Gene31	-0.53408117	
Gene02	-0.48847801	
Gene41	-0.38795085	Y
Gene06	-0.37577412	
Gene16	-0.36418134	Y
Gene26	-0.3283435	
Gene11	-0.19086011	
Gene27	-0.16533552	
Gene40	-0.15083723	
Gene42	-0.08810247	Y
Gene36	-0.0332577	
Gene34	-0.03322891	
Gene24	-0.01541293	
Gene15	-0.01315854	
Gene04	0.07310973	
Gene23	0.0832881	
Gene35	0.26230767	
Gene19	0.31809995	Y
Gene22	0.33060148	
Gene01	0.34827552	Y
Gene03	0.40059774	
Gene09	0.43365713	
Gene30	0.47885878	
Gene32	0.54061674	
Gene10	0.59040318	

Table 25 Paired Samples Statistics

	Mean	N	Std. Deviation	Std. Error Mean
Selected_Genes	-0.3951	50	1.570678	0.222127
Excluded_Genes	-0.04116	50	1.656444	0.234257

Table 26 Paired Samples Test

	Paired Differences					t	df	Significance	
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				One-Sided p	Two-Sided p
				Lower	Upper				
Selected_genes - Excluded_genes	-0.353938	0.782682	0.110688	-0.576374	-0.131502	-3.198	49	0.001	0.002

Table 27 Paired Samples Correlations

	N	Correlation	Significance	
			One-Sided p	Two-Sided p
Selected_genes & Excluded_genes	50	0.884	0	0

Table 28 Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Selected_genes	50	-2.791	4.008	-0.3951	1.570678
Excluded_genes	50	-2.439	5.58	-0.04116	1.656444

Table 29 Selected Genes: Total Variance

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	5.455	36.368	36.368	5.455	36.368	36.368	4.798	31.985	31.985
2	2.486	16.572	52.94	2.486	16.572	52.94	2.353	15.685	47.67
3	1.133	7.555	60.495	1.133	7.555	60.495	1.613	10.75	58.421
4	1.007	6.713	67.208	1.007	6.713	67.208	1.318	8.788	67.208
5	0.854	5.694	72.902						
6	0.777	5.178	78.08						
7	0.689	4.591	82.671						
8	0.491	3.276	85.947						
9	0.465	3.098	89.046						
10	0.426	2.837	91.883						
11	0.368	2.451	94.335						
12	0.302	2.015	96.35						
13	0.221	1.472	97.822						
14	0.175	1.166	98.988						
15	0.152	1.012	100						

Fig. 8 Selected Genes: Scree Plot

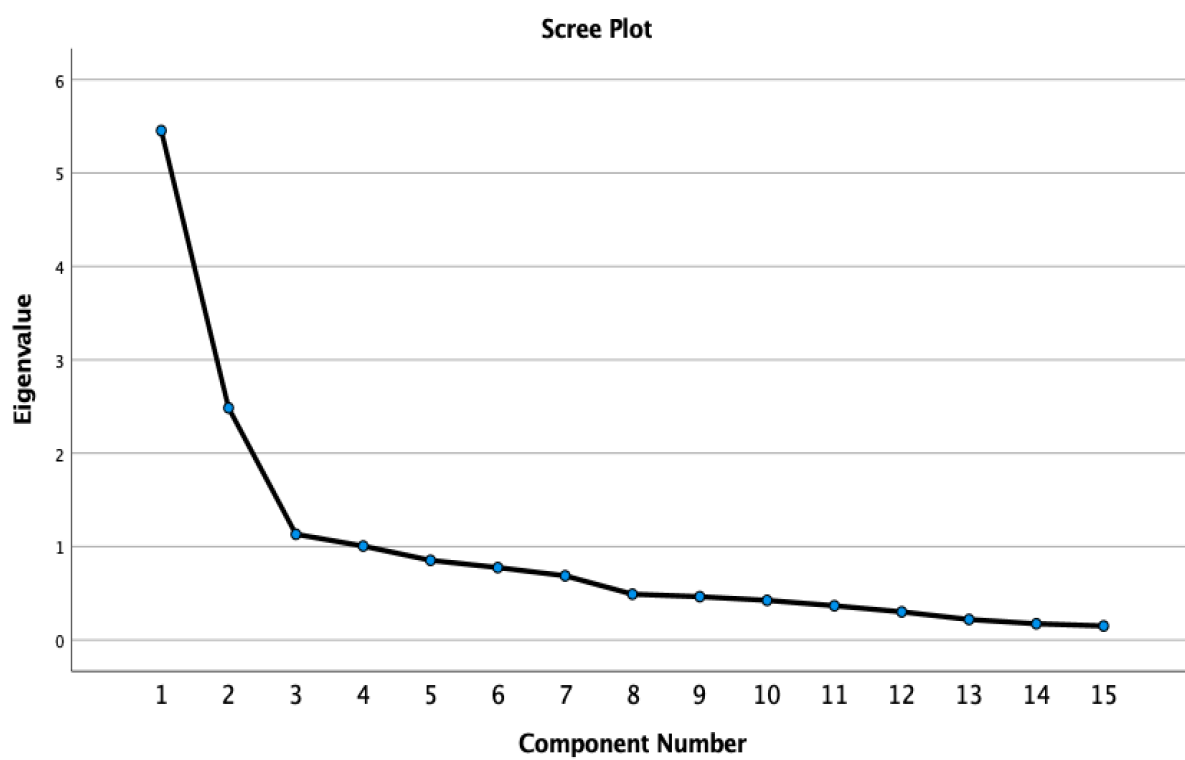


Table 30 Selected Genes: Communalities

	Initial	Extraction
Gene05	1	0.7
Gene07	1	0.8
Gene08	1	0.6
Gene12	1	0.5
Gene13	1	0.6
Gene14	1	0.7
Gene18	1	0.5
Gene20	1	0.7
Gene21	1	0.6
Gene25	1	0.8
Gene28	1	0.7
Gene29	1	0.8
Gene37	1	0.7
Gene38	1	0.8
Gene39	1	0.7
Average		0.7

Table 31 Selected Genes: Rotation Component Matrix

	Identified Genes	1	2	3	4
Gene38	TGFB3	0.824			
Gene29	MMP2	0.765			
Gene14	CDK6	0.754			
Gene13	CDK4	0.734			
Gene37	TGFB1	0.722		0.403	
Gene21	EGFR	0.705			
Gene20	EGF	0.680			0.421
Gene25	IRS2	0.651		0.573	
Gene12	CDK2	0.578			
Gene39	TGFBR2		0.787		
Gene08	CCND2		0.754		
Gene05	BID		0.625	0.485	
Gene18	CTNNB1		0.522		
Gene28	MDM2			0.825	
Gene07	CCND1				0.880

Fig. 9 Selected Genes: Component Plot in Space

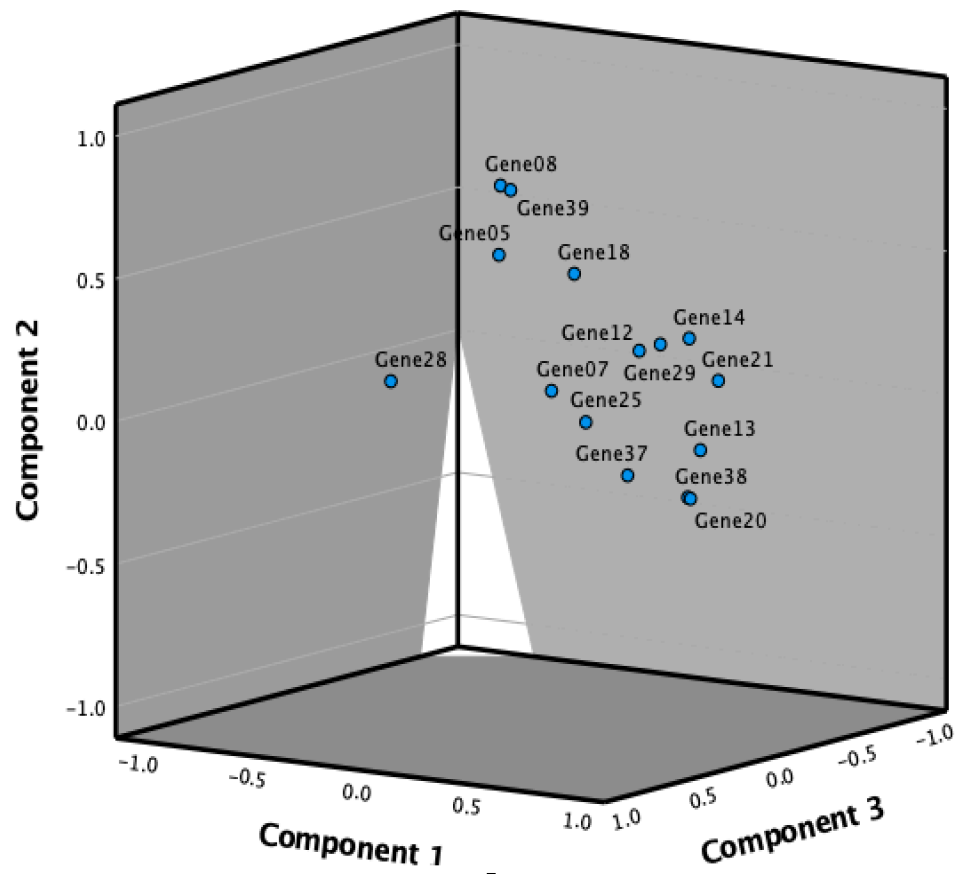


Fig. 10 Selected Genes - Derived Stimulus Configuration: Euclidean Distance Model

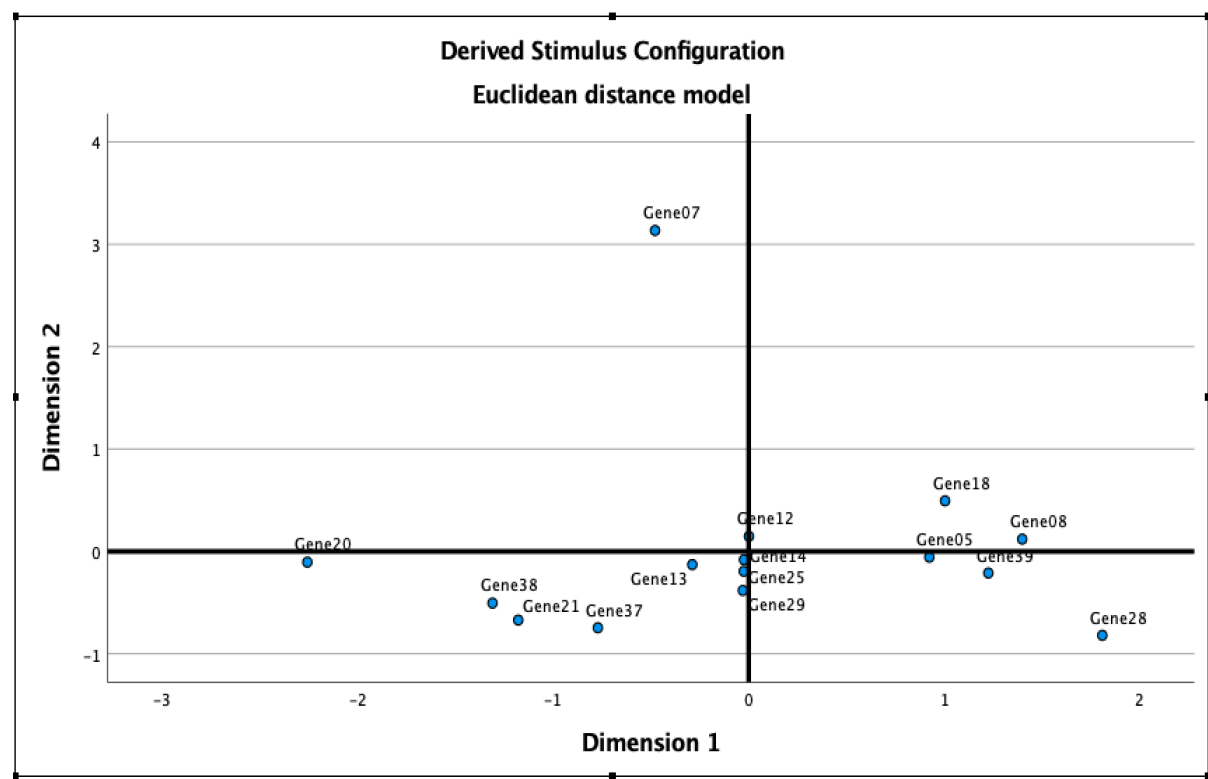


Fig. 11 Selected Genes - Scatterplot of Linear Fit: Euclidean Distance Model

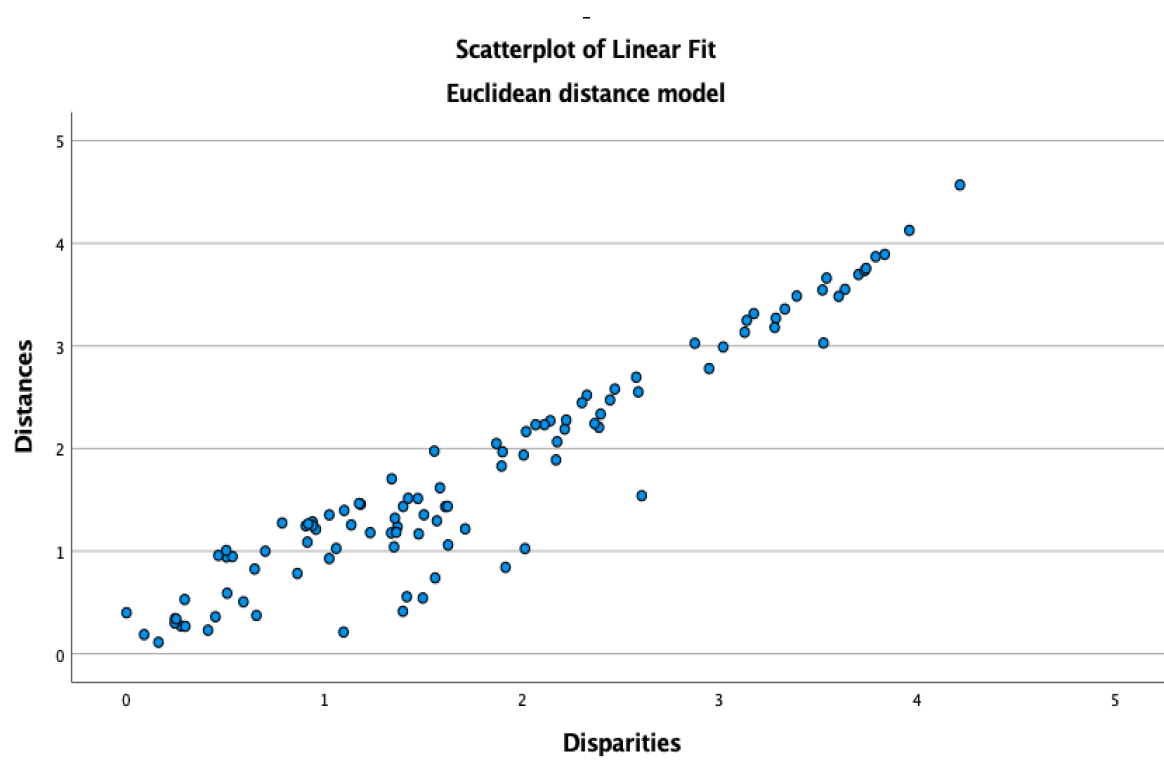
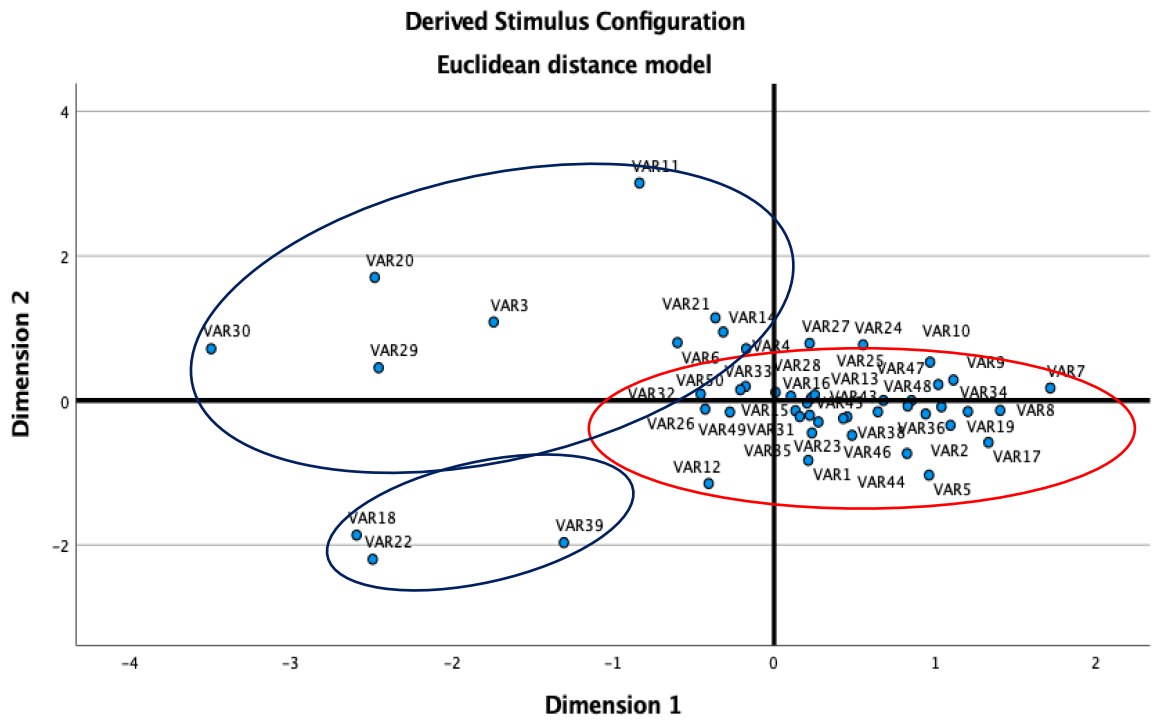


Fig. 12 Patients Associated with the Selected Genes - Derived Stimulus Configuration:
Euclidean Distance Model



Footnote: Red circle – cluster two; Blue circles – cluster one

Fig. 13 Patients Associated with the Selected Genes – Scatterplot of Linear Fit: Euclidean Distance Model

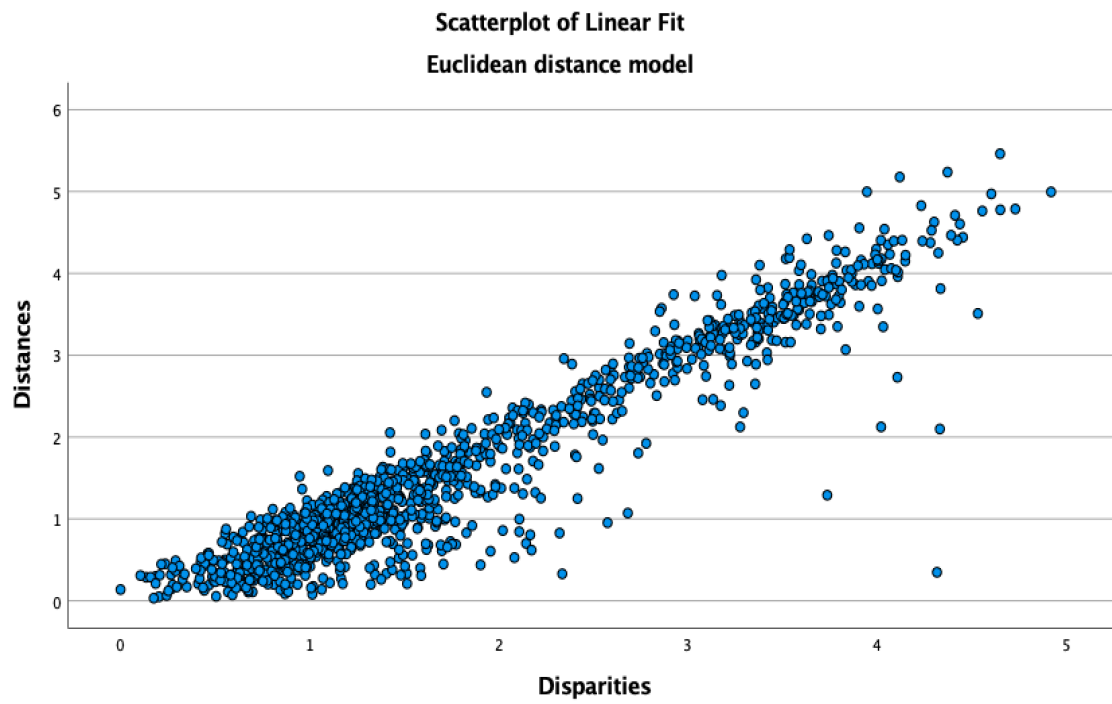


Fig 14: Selected Genes of Patients: TwoStep Cluster Analysis

Model Summary

Algorithm	TwoStep
Inputs	15
Clusters	2

Cluster Quality

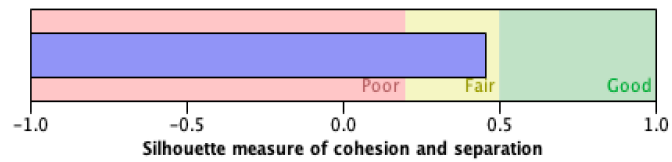
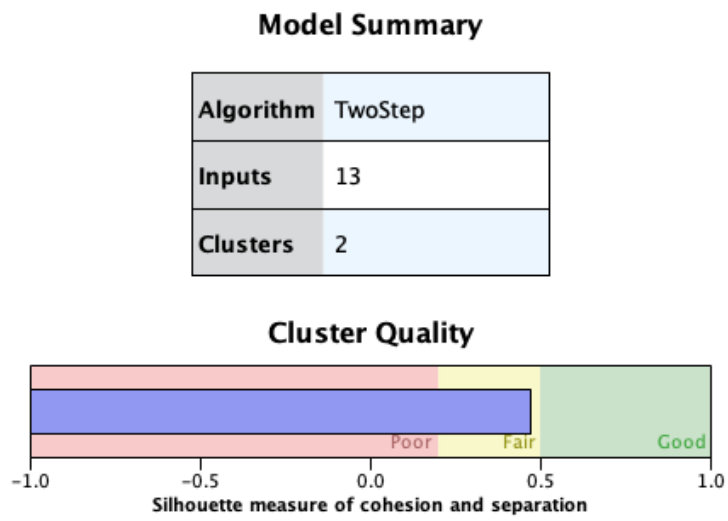


Fig. 15 Selected Genes: TwoStep Cluster Analysis



Footnote: Removed disparate genes 07 and 28

Table 32 Selected and Excluded Genes for each Patient: Clusters

Gene	Selected Genes Averaged DDCt	Excluded Genes Averaged DDCt	Gene05	Gene07	Gene08	Gene12	Gene13	Gene14	Gene18	Gene20	Gene21	Gene25	Gene28	Gene29	Gene37	Gene38	Gene39	Cluster
Patient03	1.36700	2.47500	-1.69000	0.74000	-7.43000	3.03000	2.87000	3.55000	-0.76000	7.53000	4.20000	2.69000	-4.63000	2.99000	4.23000	6.42000	-3.23000	1
Patient04	0.31600	-0.51300	0.50000	-2.41000	-2.54000	-0.15000	-0.64000	1.75000	2.15000	-3.23000	1.26000	-1.77000	-0.63000	4.57000	2.12000	2.81000	0.96000	1
Patient05	0.49800	1.43400	-2.61000	-1.38000	-3.28000	-0.51000	1.46000	1.20000	-1.38000	2.83000	3.13000	1.76000	2.62000	1.34000	3.94000	3.07000	-4.74000	1
Patient11	1.87200	0.29600	1.05000	-0.42000	-2.27000	0.43000	-2.07000	-0.91000	-0.24000	0.09000	2.10000	4.52000	20.24000	0.11000	3.15000	2.93000	-0.62000	1
Patient14	-0.04500	0.55500	-1.27000	-2.69000	-6.27000	-1.17000	-5.24000	0.51000	0.04000	3.79000	3.18000	0.63000	-1.46000	2.12000	1.27000	3.41000	2.45000	1
Patient18	2.86000	1.93400	0.88000	1.05000	4.44000	0.62000	0.52000	2.03000	19.96000	1.50000	2.12000	2.31000	1.70000	0.90000	2.10000	1.08000	1.68000	1
Patient20	3.23900	5.58000	-0.54000	-0.97000	-4.37000	4.97000	5.22000	4.41000	0.42000	10.08000	8.20000	3.91000	-3.61000	3.93000	7.58000	10.54000	-1.20000	1
Patient21	1.19800	3.07200	1.52000	-5.48000	-0.37000	0.18000	1.12000	2.18000	0.45000	-0.62000	3.96000	3.17000	1.40000	2.03000	3.39000	2.54000	2.49000	1
Patient29	4.00800	4.69100	3.34000	4.58000	4.53000	2.32000	1.94000	1.75000	3.43000	3.56000	6.85000	3.69000	2.68000	6.68000	5.75000	3.93000	5.09000	1
Patient30	3.89200	4.32300	-1.89000	7.94000	-0.04000	1.58000	5.32000	4.30000	1.94000	14.62000	7.58000	4.41000	-0.98000	5.35000	6.94000	8.73000	-7.43000	1
Patient01	-0.05500	0.27200	1.09000	1.31000	0.81000	0.21000	-0.59000	0.93000	-1.49000	-2.98000	-2.71000	1.01000	-0.11000	1.83000	-0.58000	-2.12000	2.58000	2
Patient02	-2.08900	-1.34500	-2.83000	-2.80000	-2.46000	-4.60000	-1.35000	-2.39000	-0.36000	-0.56000	-4.83000	-1.20000	-0.84000	-4.76000	2.71000	-2.42000	-2.63000	2
Patient05	-0.96200	-0.56500	-0.56000	-0.58000	0.55000	1.01000	-1.77000	-1.22000	-0.78000	-4.87000	3.65000	-1.20000	-0.99000	-1.85000	-6.96000	-1.99000	3.16000	2
Patient07	-2.36600	-0.98800	-5.17000	-4.35000	-3.86000	-1.88000	-0.22000	-1.00000	-4.54000	0.82000	8.71000	-4.93000	-3.88000	-5.52000	-3.04000	-2.25000	-4.40000	2
Patient08	-2.79100	-1.25400	-2.45000	-4.34000	-4.85000	-4.59000	-5.77000	-1.14000	-2.97000	-2.74000	-1.43000	-3.27000	-3.47000	-0.06000	0.86000	-4.80000	-2.79000	2
Patient09	-1.89400	-1.03000	-0.34000	-6.86000	-0.88000	-3.48000	-1.91000	-3.39000	-1.38000	-1.10000	0.35000	-5.09000	-2.91000	0.29000	0.58000	1.38000	-3.69000	2
Patient10	-1.20700	-0.30700	1.19000	-7.76000	-0.21000	-2.39000	-0.85000	-2.00000	-1.47000	-1.50000	-0.72000	0.87000	0.00000	-0.32000	0.94000	-0.50000	-1.37000	2
Patient12	-0.17700	-0.12200	-0.84000	6.87000	2.06000	2.25000	-1.40000	-2.35000	-0.31000	0.79000	-1.22000	-0.98000	-1.52000	-2.93000	-0.21000	1.16000	-4.02000	2
Patient13	-1.66700	-0.82500	-2.60000	-3.13000	-3.02000	-2.36000	-3.11000	-2.77000	-1.22000	-0.28000	0.41000	-2.15000	-1.58000	2.22000	-1.58000	-1.91000	-1.60000	2
Patient15	-0.42600	0.19900	-0.35000	0.27000	-2.31000	0.34000	-1.29000	-0.58000	-0.55000	-0.80000	-0.01000	1.39000	0.75000	-1.76000	0.42000	-2.20000	0.29000	2
Patient16	-0.74700	0.25900	2.08000	-1.82000	-3.07000	0.87000	2.26000	-3.49000	1.01000	0.80000	-0.35000	-1.54000	-1.79000	-0.05000	0.04000	0.35000	-0.32000	2
Patient17	-1.73800	-2.43900	1.31000	-1.88000	2.73000	0.41000	-1.87000	-2.77000	-2.62000	-5.56000	-0.66000	-2.97000	0.36000	-1.17000	-2.94000	-5.18000	-0.63000	2
Patient19	-2.06000	1.40600	2.39000	-4.15000	0.34000	-0.38000	-0.15000	-2.02000	-3.30000	-4.39000	-3.64000	-3.46000	-1.48000	-2.17000	-0.77000	5.76000	2.18000	2
Patient22	1.47000	0.41700	1.40000	20.28000	-3.86000	0.20000	-1.27000	-2.82000	1.07000	6.32000	1.80000	1.16000	-2.62000	-1.88000	0.72000	5.13000	-0.76000	2
Patient23	0.94500	0.14600	2.09000	0.35000	1.94000	2.64000	0.68000	-2.30000	1.20000	0.40000	-0.27000	-0.39000	-1.04000	-3.36000	3.11000	-2.18000	-0.51000	2
Patient24	0.61700	0.58400	0.45000	-7.76000	-0.63000	-3.41000	-1.21000	1.53000	-1.16000	-0.69000	0.34000	0.99000	-0.75000	-0.26000	1.68000	1.86000	0.66000	2
Patient25	1.75600	-1.75500	-2.12000	-4.43000	-0.82000	-1.22000	-2.24000	-0.33000	-2.55000	0.37000	1.38000	-1.41000	-1.64000	-5.57000	-1.93000	-0.51000	-3.32000	2
Patient26	0.51600	0.25000	-0.79000	1.06000	1.53000	0.67000	-2.45000	0.86000	-0.05000	0.09000	2.38000	-1.22000	1.47000	1.73000	4.76000	-2.78000	0.48000	2
Patient27	-0.58900	-1.13900	-2.48000	-3.22000	-2.29000	3.47000	-0.55000	-3.98000	-3.18000	2.75000	-4.05000	0.42000	0.49000	-0.02000	1.62000	3.94000	-1.26000	2
Patient28	0.00600	0.72300	-0.56000	-1.51000	0.43000	2.13000	0.81000	1.10000	-1.09000	1.52000	-1.75000	-2.27000	1.90000	-1.44000	0.43000	0.46000	-0.08000	2
Patient31	-0.69500	0.03600	-0.85000	-1.10000	1.45000	-1.35000	-2.40000	-1.37000	-2.08000	0.94000	1.51000	-2.09000	-3.91000	-0.72000	-0.05000	1.61000	0.00000	2
Patient32	0.01500	-0.23500	-2.53000	-1.04000	0.09000	-2.19000	2.22000	-1.28000	-0.71000	6.86000	0.01000	1.24000	-1.76000	-1.60000	0.26000	1.56000	-0.91000	2
Patient33	-0.29200	1.04300	-1.71000	-2.27000	-1.31000	0.38000	0.16000	-2.27000	-1.53000	5.08000	0.76000	1.20000	-2.41000	-0.81000	0.15000	1.36000	-1.14000	2
Patient34	-1.76400	-1.59900	3.85000	-3.70000	-2.03000	-2.66000	-3.06000	-2.26000	-2.78000	-1.16000	0.41000	-2.94000	-1.26000	-1.59000	-3.67000	-0.60000	-3.00000	2
Patient35	-1.07300	-0.91800	-1.51000	1.29000	-2.02000	-1.58000	-2.74000	0.81000	-1.89000	1.90000	1.97000	-3.63000	0.74000	-3.81000	-1.85000	-1.13000	-2.65000	2
Patient36	-2.15400	-2.02100	-5.27000	-2.32000	-2.43000	-1.87000	-0.26000	-2.04000	-3.91000	-2.20000	1.28000	-3.20000	-1.76000	-1.83000	-2.26000	-1.76000	-2.45000	2
Patient37	-0.72000	-0.62900	-1.89000	-2.86000	-2.70000	-0.06000	1.67000	-0.49000	-0.33000	-1.93000	0.26000	-0.08000	-2.08000	-0.25000	-0.63000	0.59000	-0.03000	2
Patient38	-1.33000	-1.07300	-1.57000	-1.43000	-2.25000	-2.08000	-2.00000	-1.64000	-0.73000	-2.68000	-1.12000	-1.09000	-3.06000	-0.54000	0.63000	-0.11000	-0.27000	2
Patient39	0.63800	-0.40600	-0.62000	16.10000	-1.31000	-0.44000	-0.30000	-0.26000	0.16000	0.31000	0.47000	-0.90000	-0.55000	-0.98000	-0.53000	-0.39000	-1.20000	2
Patient40	-0.81400	-0.05100	-2.87000	-0.74000	-0.17000	-1.39000	-1.19000	-1.21000	-1.25000	0.35000	1.77000	-3.94000	-0.69000	-0.69000	-0.85000	0.16000	0.18000	2
Patient41	-0.56900	-0.05000	1.49000	-2.25000	-0.36000	-0.76000	-1.36000	-0.59000	-1.09000	-0.45000	1.04000	-1.74000	-1.64000	0.64000	0.79000	1.64000	-0.91000	2
Patient42	-0.75200	-0.69800	-2.75000	-2.53000	0.51000	-2.43000	-0.62000	-0.50000	-0.30000	-1.08000	0.37000	-0.15000	-2.41000	-0.69000	0.47000	1.55000	-0.73000	2
Patient43	-1.54500	-1.35300	-3.48000	-3.45000	0.25000	-2.52000	-2.92000	-0.87000	-2.47000	1.15000	2.40000	-2.57000	-3.50000	-2.30000	-1.07000	-0.96000	-0.87000	2
Patient44	-1.57200	-1.92100	-3.70000	-3.61000	-0.44000	-2.17000	-2.70000	-2.62000	-1.93000	9.16000	-2.06000	-3.60000	-1.73000	-2.88000	-1.89000	-1.84000	-1.56000	2
Patient45	-1.04500	-0.99600	-2.56000	-1.95000	0.47000	-2.04000	-1.72000	-1.88000	0.34000	-2.04000	-0.73000	-2.12000	-0.87000	-1.16000	-0.93000	0.82000	0.72000	2
Patient46	-1.32600	-1.19900	-2.54000	0.98000	-0.34000	-3.21000	-3.13000	0.90000	-2.60000	-0.90000	-0.87000	-0.01000	0.50000	-2.23000	-0.53000	-3.38000	-2.54000	2
Patient47	-2.24100	-0.55100	-6.52000	-4.16000	-3.55000	-3.46000	-0.75000	-3.52000	-1.54000	-0.88000	-0.44000	-2.25000	-4.69000	-2.16000	0.69000	4.13000	-4.53000	2
Patient48	-1.87300	-1.19300	-2.98000	-2.44000	-2.46000	-1.31000	-1.43000	-2.81000	-3.50000	-3.18000	-1.89000	-2.03000	-3.04000	-0.31000	3.07000	-1.35000	-2.44000	2
Patient49	0.11900	-0.19800	-2.15000	0.13000	1.12000	1.53000	0.58000	-2.25000	-0.19000	3.33000	0.04000	-4.02000	0.61000	-1.90000	1.74000	2.16000	1.07000	2
Patient50	0.12400	0.11700	-0.80000	-1.28000	0.36000	-1.35000	-2.02000	0.35000	-1.49000	3.24000	2.04000	-0.54000	0.10000	-0.46000	2.57000	0.44000	0.69000	2

Fig. 16 Concentration Ellipse

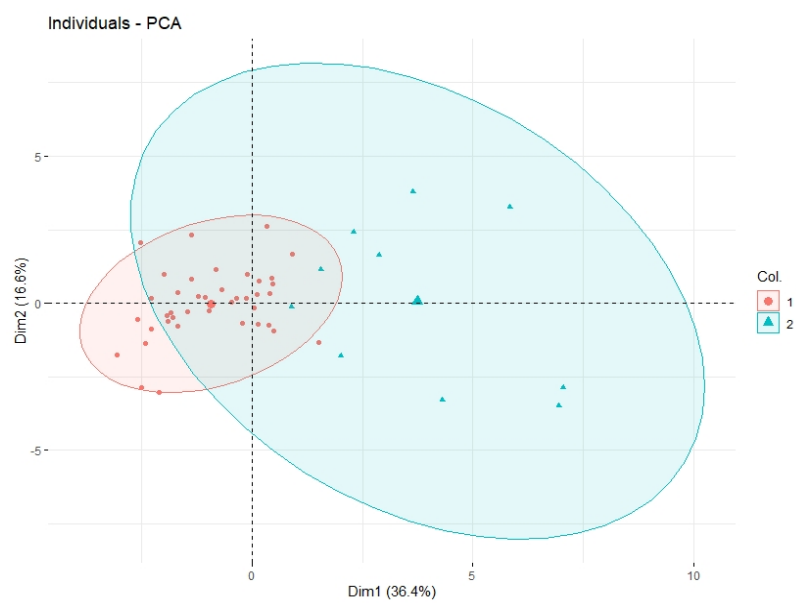


Fig. 17 Confidence Ellipse

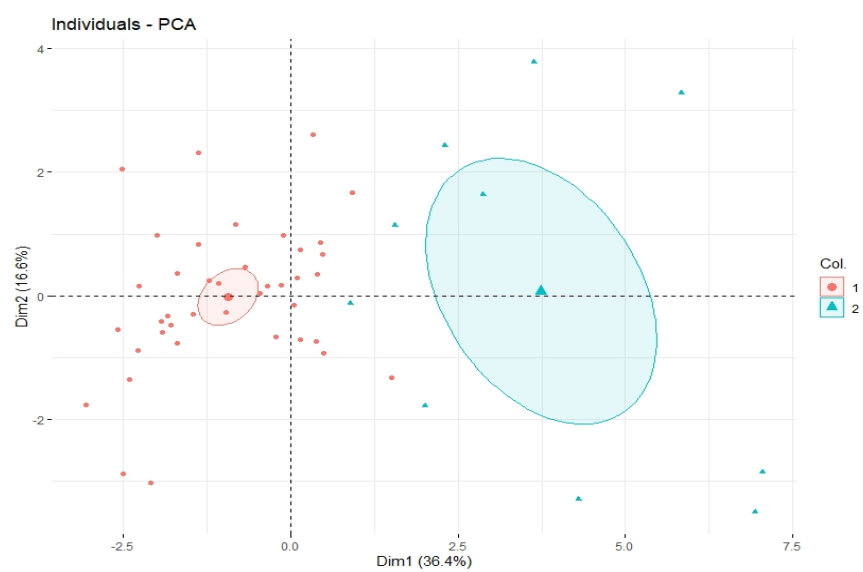


Fig. 18 Patient Analysis

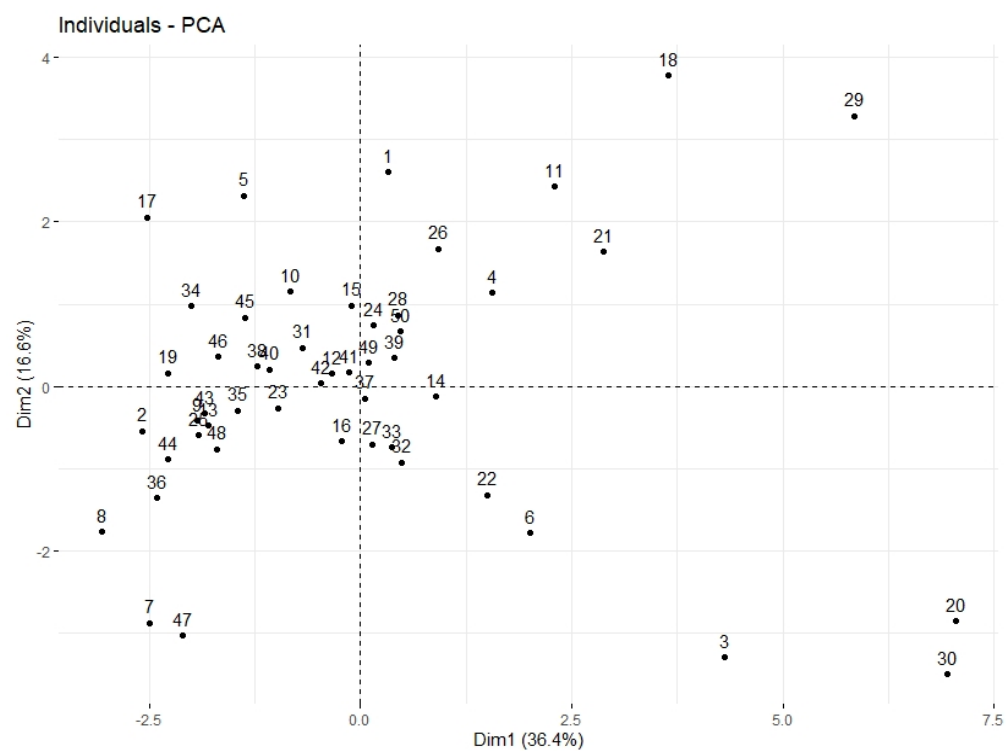


Fig. 19

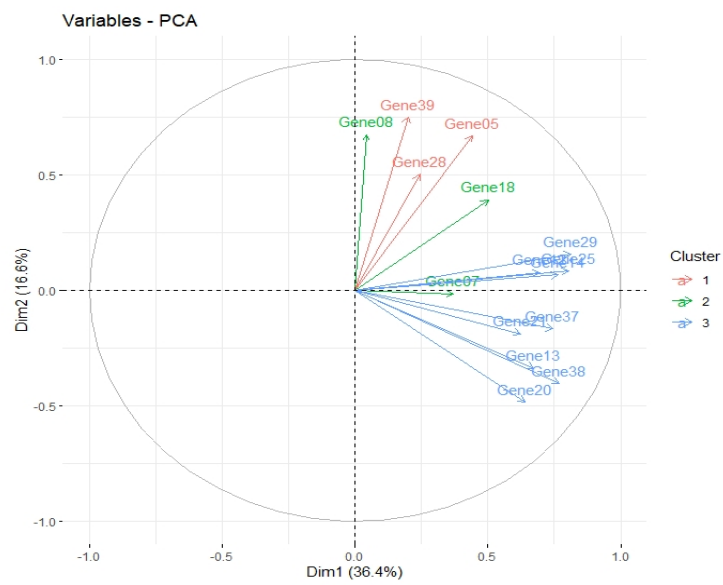


Fig. 20 PCA Biplot

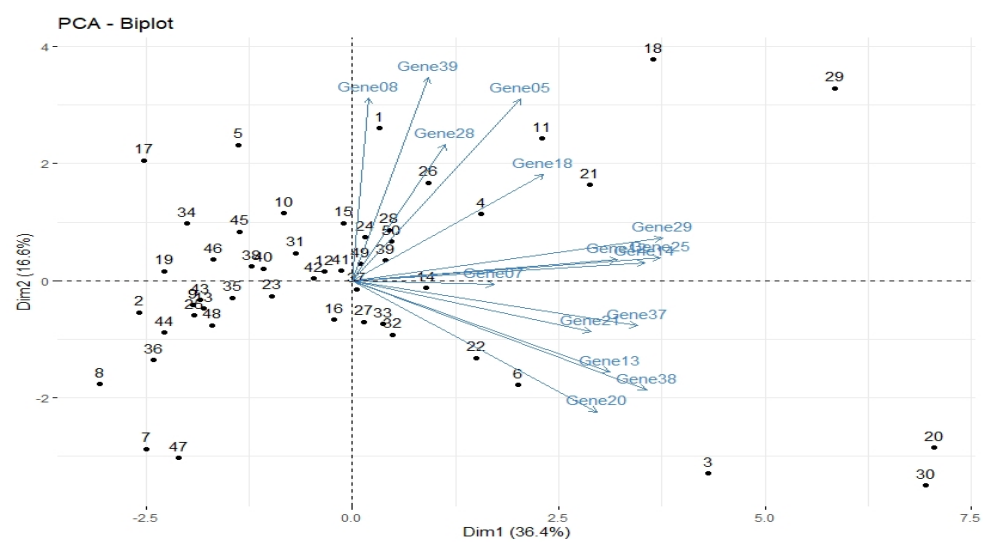


Fig. 21 Contribution of Variables to Component One

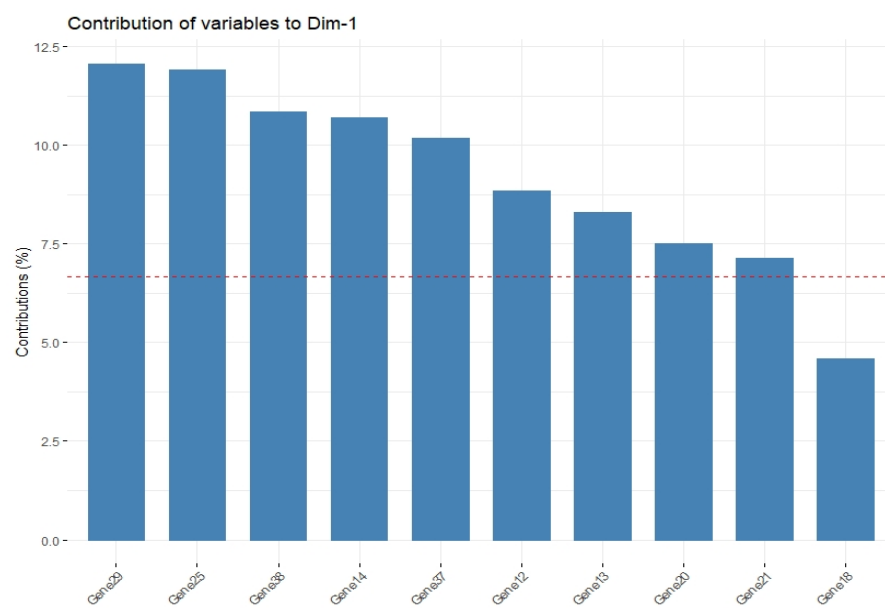


Fig. 22 Contribution of Variables to Component Two

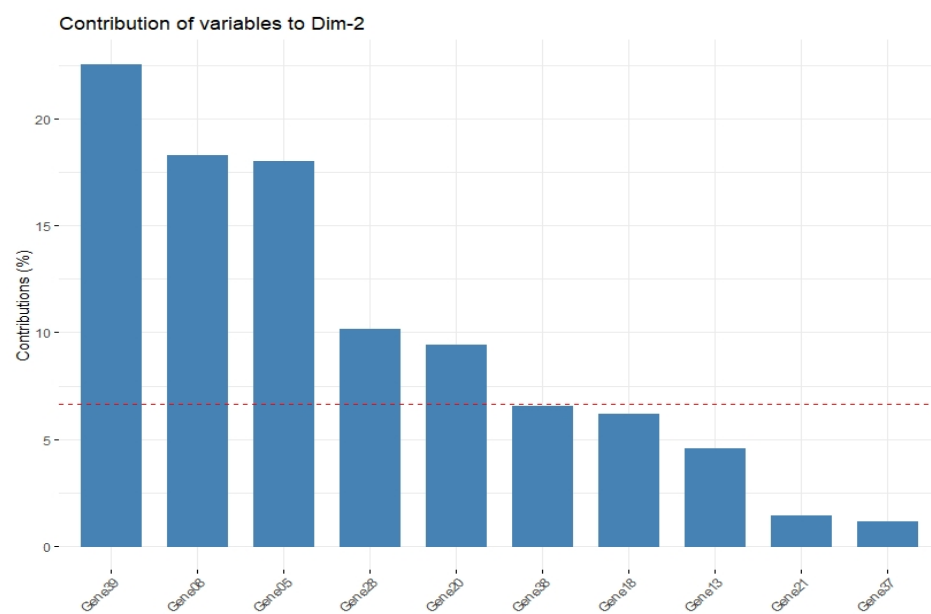


Fig. 23 Contribution of Variables to Components One and Two

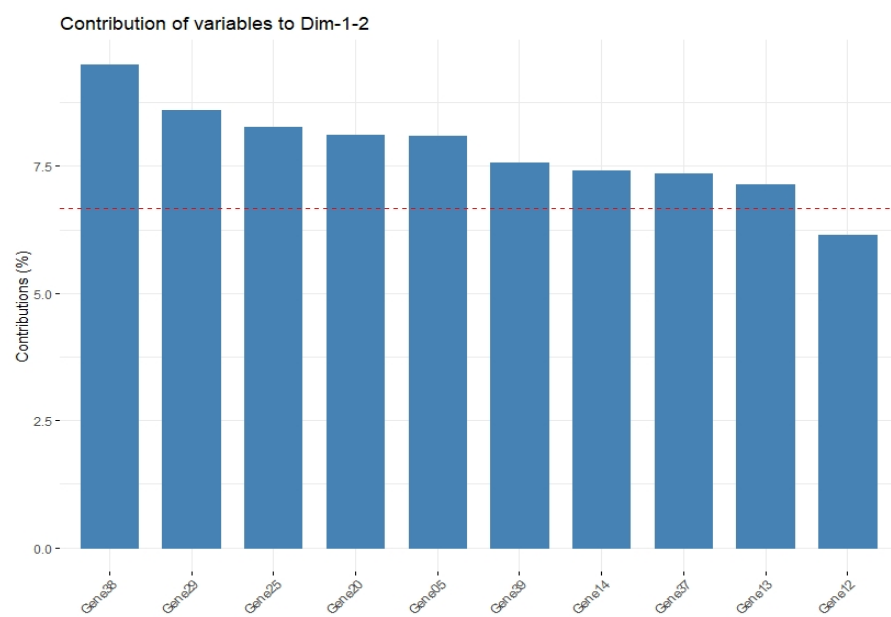


Table 33 Pairwise T-Test (p adjusted Bonferroni method)

ID	.y.	Group1	Group2	n1	n2	p.adj	p.adj.signif
12	Values	BID	TGFB1	50	50	0.000235	***
63	Values	CDK6	EGFR	50	50	0.000364	***
8	Values	BID	EGFR	50	50	0.00065	***
13	Values	BID	TGFB3	50	50	0.001	**
93	Values	IRS2	TGFB1	50	50	0.002	**
67	Values	CDK6	TGFB1	50	50	0.004	**
100	Values	MMP2	TGFB1	50	50	0.004	**
58	Values	CDK4	TGFB1	50	50	0.009	**
59	Values	CDK4	TGFB3	50	50	0.009	**
94	Values	IRS2	TGFB3	50	50	0.015	*
101	Values	MMP2	TGFB3	50	50	0.019	*
68	Values	CDK6	TGFB3	50	50	0.02	*
49	Values	CDK2	TGFB3	50	50	0.023	*
44	Values	CDK2	EGFR	50	50	0.031	*
48	Values	CDK2	TGFB1	50	50	0.037	*
54	Values	CDK4	EGFR	50	50	0.04	*
90	Values	EGFR	TGFBR2	50	50	0.055	ns
7	Values	BID	EGF	50	50	0.058	ns
33	Values	CCND2	EGFR	50	50	0.066	ns
87	Values	EGFR	MMP2	50	50	0.066	ns
37	Values	CCND2	TGFB1	50	50	0.074	ns
104	Values	TGFB1	TGFBR2	50	50	0.074	ns
85	Values	EGFR	IRS2	50	50	0.083	ns
105	Values	TGFB3	TGFBR2	50	50	0.204	ns
62	Values	CDK6	EGF	50	50	0.205	ns
38	Values	CCND2	TGFB3	50	50	0.268	ns
53	Values	CDK4	EGF	50	50	0.311	ns
43	Values	CDK2	EGF	50	50	0.417	ns
79	Values	EGF	IRS2	50	50	0.425	ns
32	Values	CCND2	EGF	50	50	0.623	ns
75	Values	CTNNB1	TGFB1	50	50	0.708	ns
71	Values	CTNNB1	EGFR	50	50	0.954	ns
1	Values	BID	CCND1	50	50	1	ns

2	Values	BID	CCND2	50	50	1	ns
3	Values	BID	CDK2	50	50	1	ns
4	Values	BID	CDK4	50	50	1	ns
5	Values	BID	CDK6	50	50	1	ns
6	Values	BID	CTNNB1	50	50	1	ns
9	Values	BID	IRS2	50	50	1	ns
10	Values	BID	MDM2	50	50	1	ns
11	Values	BID	MMP2	50	50	1	ns
14	Values	BID	TGFBR2	50	50	1	ns
15	Values	CCND1	CCND2	50	50	1	ns
16	Values	CCND1	CDK2	50	50	1	ns
17	Values	CCND1	CDK4	50	50	1	ns
18	Values	CCND1	CDK6	50	50	1	ns
19	Values	CCND1	CTNNB1	50	50	1	ns
20	Values	CCND1	EGF	50	50	1	ns
21	Values	CCND1	EGFR	50	50	1	ns
22	Values	CCND1	IRS2	50	50	1	ns
23	Values	CCND1	MDM2	50	50	1	ns
24	Values	CCND1	MMP2	50	50	1	ns
25	Values	CCND1	TGFB1	50	50	1	ns
26	Values	CCND1	TGFB3	50	50	1	ns
27	Values	CCND1	TGFBR2	50	50	1	ns
28	Values	CCND2	CDK2	50	50	1	ns
29	Values	CCND2	CDK4	50	50	1	ns
30	Values	CCND2	CDK6	50	50	1	ns
31	Values	CCND2	CTNNB1	50	50	1	ns
34	Values	CCND2	IRS2	50	50	1	ns
35	Values	CCND2	MDM2	50	50	1	ns
36	Values	CCND2	MMP2	50	50	1	ns
39	Values	CCND2	TGFBR2	50	50	1	ns
40	Values	CDK2	CDK4	50	50	1	ns
41	Values	CDK2	CDK6	50	50	1	ns
42	Values	CDK2	CTNNB1	50	50	1	ns
45	Values	CDK2	IRS2	50	50	1	ns
46	Values	CDK2	MDM2	50	50	1	ns
47	Values	CDK2	MMP2	50	50	1	ns

50	Values	CDK2	TGFBR2	50	50	1	ns
51	Values	CDK4	CDK6	50	50	1	ns
52	Values	CDK4	CTNNB1	50	50	1	ns
55	Values	CDK4	IRS2	50	50	1	ns
56	Values	CDK4	MDM2	50	50	1	ns
57	Values	CDK4	MMP2	50	50	1	ns
60	Values	CDK4	TGFBR2	50	50	1	ns
61	Values	CDK6	CTNNB1	50	50	1	ns
64	Values	CDK6	IRS2	50	50	1	ns
65	Values	CDK6	MDM2	50	50	1	ns
66	Values	CDK6	MMP2	50	50	1	ns
69	Values	CDK6	TGFBR2	50	50	1	ns
70	Values	CTNNB1	EGF	50	50	1	ns
72	Values	CTNNB1	IRS2	50	50	1	ns
73	Values	CTNNB1	MDM2	50	50	1	ns
74	Values	CTNNB1	MMP2	50	50	1	ns
76	Values	CTNNB1	TGFB3	50	50	1	ns
77	Values	CTNNB1	TGFBR2	50	50	1	ns
78	Values	EGF	EGFR	50	50	1	ns
80	Values	EGF	MDM2	50	50	1	ns
81	Values	EGF	MMP2	50	50	1	ns
82	Values	EGF	TGFB1	50	50	1	ns
83	Values	EGF	TGFB3	50	50	1	ns
84	Values	EGF	TGFBR2	50	50	1	ns
86	Values	EGFR	MDM2	50	50	1	ns
88	Values	EGFR	TGFB1	50	50	1	ns
89	Values	EGFR	TGFB3	50	50	1	ns
91	Values	IRS2	MDM2	50	50	1	ns
92	Values	IRS2	MMP2	50	50	1	ns
95	Values	IRS2	TGFBR2	50	50	1	ns
96	Values	MDM2	MMP2	50	50	1	ns
97	Values	MDM2	TGFB1	50	50	1	ns
98	Values	MDM2	TGFB3	50	50	1	ns
99	Values	MDM2	TGFBR2	50	50	1	ns
102	Values	MMP2	TGFBR2	50	50	1	ns
103	Values	TGFB1	TGFB3	50	50	1	ns

Table 34 Pairwise T-Test (p adjusted BH method)

ID	.y.	Group1	Group2	n1	n2	p.adj	p.adj.signif
12	Values	BID	TGFB1	50	50	0.000182	***
63	Values	CDK6	EGFR	50	50	0.000182	***
8	Values	BID	EGFR	50	50	0.000217	***
13	Values	BID	TGFB3	50	50	0.000373	***
93	Values	IRS2	TGFB1	50	50	0.000426	***
67	Values	CDK6	TGFB1	50	50	0.000532	***
100	Values	MMP2	TGFB1	50	50	0.000532	***
58	Values	CDK4	TGFB1	50	50	0.001	**
59	Values	CDK4	TGFB3	50	50	0.001	**
94	Values	IRS2	TGFB3	50	50	0.001	**
44	Values	CDK2	EGFR	50	50	0.002	**
48	Values	CDK2	TGFB1	50	50	0.002	**
49	Values	CDK2	TGFB3	50	50	0.002	**
54	Values	CDK4	EGFR	50	50	0.002	**
68	Values	CDK6	TGFB3	50	50	0.002	**
101	Values	MMP2	TGFB3	50	50	0.002	**
7	Values	BID	EGF	50	50	0.003	**
33	Values	CCND2	EGFR	50	50	0.003	**
37	Values	CCND2	TGFB1	50	50	0.003	**
87	Values	EGFR	MMP2	50	50	0.003	**
90	Values	EGFR	TGFBR2	50	50	0.003	**
104	Values	TGFB1	TGFBR2	50	50	0.003	**
85	Values	EGFR	IRS2	50	50	0.004	**
62	Values	CDK6	EGF	50	50	0.008	**
105	Values	TGFB3	TGFBR2	50	50	0.008	**
38	Values	CCND2	TGFB3	50	50	0.01	*
53	Values	CDK4	EGF	50	50	0.012	*
43	Values	CDK2	EGF	50	50	0.015	*
79	Values	EGF	IRS2	50	50	0.015	*
32	Values	CCND2	EGF	50	50	0.021	*
75	Values	CTNNB1	TGFB1	50	50	0.023	*
71	Values	CTNNB1	EGFR	50	50	0.03	*
84	Values	EGF	TGFBR2	50	50	0.031	*

76	Values	CTNNB1	TGFB3	50	50	0.032	*
81	Values	EGF	MMP2	50	50	0.032	*
97	Values	MDM2	TGFB1	50	50	0.032	*
11	Values	BID	MMP2	50	50	0.033	*
86	Values	EGFR	MDM2	50	50	0.043	*
9	Values	BID	IRS2	50	50	0.057	ns
20	Values	CCND1	EGF	50	50	0.06	ns
21	Values	CCND1	EGFR	50	50	0.06	ns
98	Values	MDM2	TGFB3	50	50	0.06	ns
3	Values	BID	CDK2	50	50	0.061	ns
26	Values	CCND1	TGFB3	50	50	0.067	ns
70	Values	CTNNB1	EGF	50	50	0.067	ns
25	Values	CCND1	TGFB1	50	50	0.077	ns
4	Values	BID	CDK4	50	50	0.097	ns
80	Values	EGF	MDM2	50	50	0.099	ns
5	Values	BID	CDK6	50	50	0.102	ns
6	Values	BID	CTNNB1	50	50	0.158	ns
10	Values	BID	MDM2	50	50	0.233	ns
14	Values	BID	TGFBR2	50	50	0.335	ns
36	Values	CCND2	MMP2	50	50	0.553	ns
102	Values	MMP2	TGFBR2	50	50	0.572	ns
29	Values	CCND2	CDK4	50	50	0.586	ns
31	Values	CCND2	CTNNB1	50	50	0.591	ns
28	Values	CCND2	CDK2	50	50	0.593	ns
2	Values	BID	CCND2	50	50	0.619	ns
34	Values	CCND2	IRS2	50	50	0.642	ns
1	Values	BID	CCND1	50	50	0.675	ns
50	Values	CDK2	TGFBR2	50	50	0.675	ns
95	Values	IRS2	TGFBR2	50	50	0.683	ns
60	Values	CDK4	TGFBR2	50	50	0.697	ns
77	Values	CTNNB1	TGFBR2	50	50	0.697	ns
30	Values	CCND2	CDK6	50	50	0.729	ns
35	Values	CCND2	MDM2	50	50	0.729	ns
66	Values	CDK6	MMP2	50	50	0.823	ns
69	Values	CDK6	TGFBR2	50	50	0.857	ns
99	Values	MDM2	TGFBR2	50	50	0.857	ns

15	Values	CCND1	CCND2	50	50	0.978	ns
16	Values	CCND1	CDK2	50	50	0.978	ns
17	Values	CCND1	CDK4	50	50	0.978	ns
18	Values	CCND1	CDK6	50	50	0.978	ns
19	Values	CCND1	CTNNB1	50	50	0.978	ns
22	Values	CCND1	IRS2	50	50	0.978	ns
23	Values	CCND1	MDM2	50	50	0.978	ns
24	Values	CCND1	MMP2	50	50	0.978	ns
27	Values	CCND1	TGFBR2	50	50	0.978	ns
39	Values	CCND2	TGFBR2	50	50	0.978	ns
40	Values	CDK2	CDK4	50	50	0.978	ns
41	Values	CDK2	CDK6	50	50	0.978	ns
42	Values	CDK2	CTNNB1	50	50	0.978	ns
45	Values	CDK2	IRS2	50	50	0.978	ns
46	Values	CDK2	MDM2	50	50	0.978	ns
47	Values	CDK2	MMP2	50	50	0.978	ns
51	Values	CDK4	CDK6	50	50	0.978	ns
52	Values	CDK4	CTNNB1	50	50	0.978	ns
55	Values	CDK4	IRS2	50	50	0.978	ns
56	Values	CDK4	MDM2	50	50	0.978	ns
57	Values	CDK4	MMP2	50	50	0.978	ns
61	Values	CDK6	CTNNB1	50	50	0.978	ns
64	Values	CDK6	IRS2	50	50	0.978	ns
65	Values	CDK6	MDM2	50	50	0.978	ns
72	Values	CTNNB1	IRS2	50	50	0.978	ns
73	Values	CTNNB1	MDM2	50	50	0.978	ns
74	Values	CTNNB1	MMP2	50	50	0.978	ns
78	Values	EGF	EGFR	50	50	0.978	ns
82	Values	EGF	TGFB1	50	50	0.978	ns
83	Values	EGF	TGFB3	50	50	0.978	ns
88	Values	EGFR	TGFB1	50	50	0.978	ns
89	Values	EGFR	TGFB3	50	50	0.978	ns
91	Values	IRS2	MDM2	50	50	0.978	ns
92	Values	IRS2	MMP2	50	50	0.978	ns
96	Values	MDM2	MMP2	50	50	0.978	ns
103	Values	TGFB1	TGFB3	50	50	0.978	ns

Table 35 P adjusted Mean for Volcano Plot

ID	Genes	N	Padj (mean)	Fold Change	Biomarker
1	BID	14	0.718563214	-1.463222396	Gene05
2	CCND1	13	1.000000000	-0.84294231	Gene07
3	CCND2	12	0.752583333	-1.122190935	Gene08
4	CDK2	11	0.682545455	-0.701018938	Gene12
5	CDK4	10	0.636900000	-0.656190577	Gene13
6	CDK6	9	0.581040444	-0.796383848	Gene14
7	CTNNB1	8	0.957750000	-0.642789546	Gene18
8	EGF	7	0.516285714	0.856166215	Gene20
9	EGFR	8	0.386501750	0.878026455	Gene21
10	IRS2	9	0.834222222	-0.671274135	Gene25
11	MDM2	10	1.000000000	-0.71709458	Gene28
12	MMP2	11	0.915090909	-0.609917239	Gene29
13	TGFB1	12	0.403186250	0.786919468	Gene37
14	TGFB3	13	0.488846154	0.806230549	Gene38
15	TGFBR2	14	0.809500000	-1.030795501	Gene39

Fig. 24 Volcano Plot

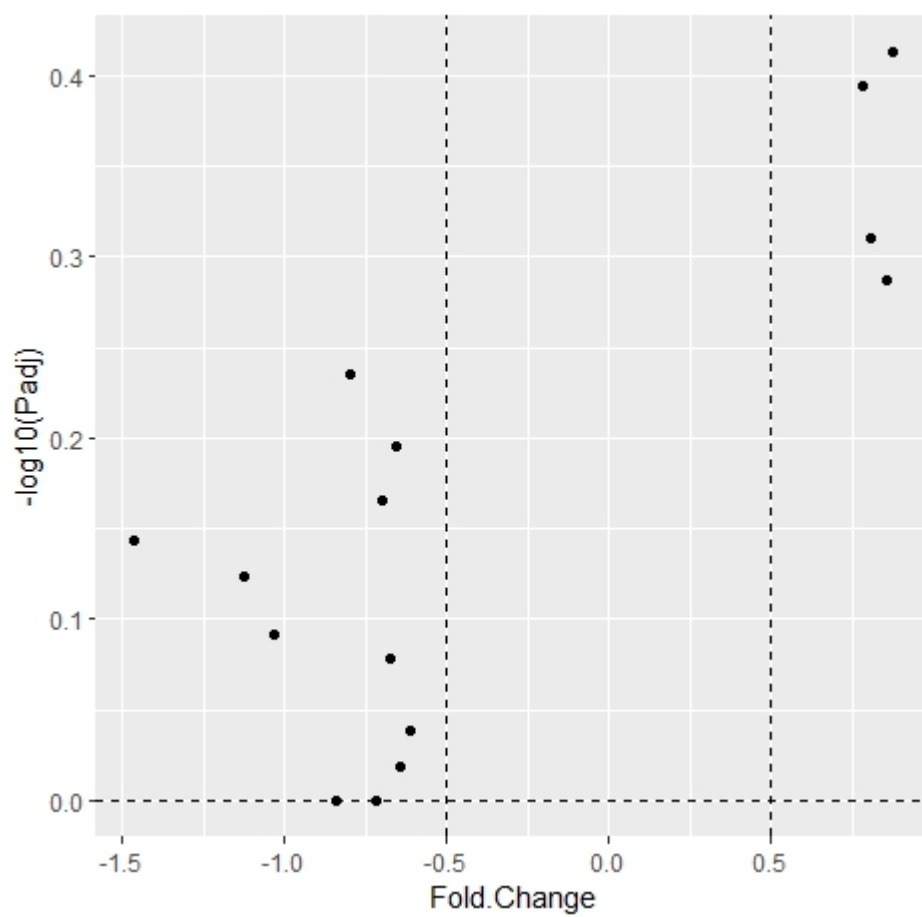


Table 36 Significant Predictors: Model 1

	B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I. for EXP(B)	
							Lower	Upper
Gene03	0.49	0.22	5.05	1.00	0.03	1.63	1.07	2.50
Gene05	-0.48	0.21	5.01	1.00	0.03	0.62	0.41	0.94
Gene08	-0.28	0.15	3.56	1.00	0.06	0.76	0.57	1.01
Gene11	0.36	0.17	4.45	1.00	0.04	1.44	1.03	2.01
Gene14	-0.73	0.33	5.00	1.00	0.03	0.48	0.25	0.91
Gene15	0.53	0.28	3.65	1.00	0.06	1.70	0.99	2.93
Gene21	0.27	0.15	3.14	1.00	0.08	1.31	0.97	1.76
Gene23	0.30	0.18	2.72	1.00	0.10	1.35	0.94	1.94
Gene25	-0.73	0.26	7.73	1.00	0.01	0.48	0.29	0.81
Gene31	-0.28	0.15	3.63	1.00	0.06	0.76	0.57	1.01
Gene37	0.56	0.21	6.90	1.00	0.01	1.75	1.15	2.65

Fig. 26 Significant Predictors: Model 1: A summary of the gene expression levels of healthy (0) compared to non-healthy patients (1)

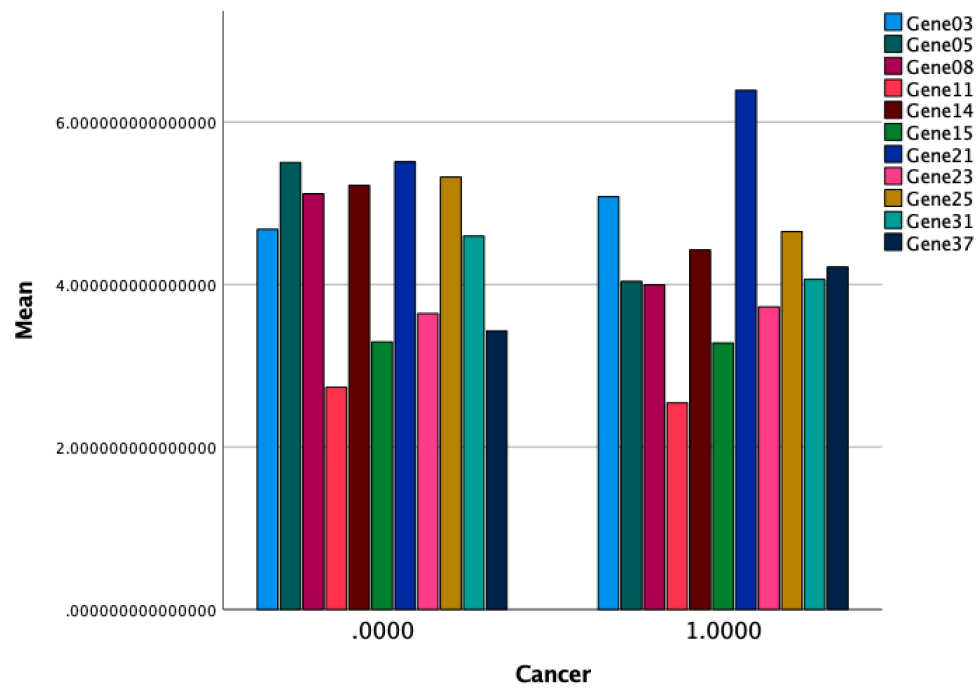


Table 37 Logistic Regression (Overall Prediction Accuracy: Model 1)

		Predicted Cancer		Percentage Correct
		Healthy (0)	Non-Healthy (1)	
Observed Cancer	Healthy (0)	41	9	82
	Non-Healthy (1)	11	39	78
Overall Percentage				80

Table 38 Area Under the Curve: Model 1

Test Result Variable(s)	Area
Gene03	0.526
Gene05	0.33
Gene08	0.372
Gene11	0.454
Gene14	0.361
Gene15	0.478
Gene21	0.604
Gene23	0.492
Gene25	0.387
Gene31	0.384
Gene37	0.574

Fig. 27 ROC Curve: Model 2

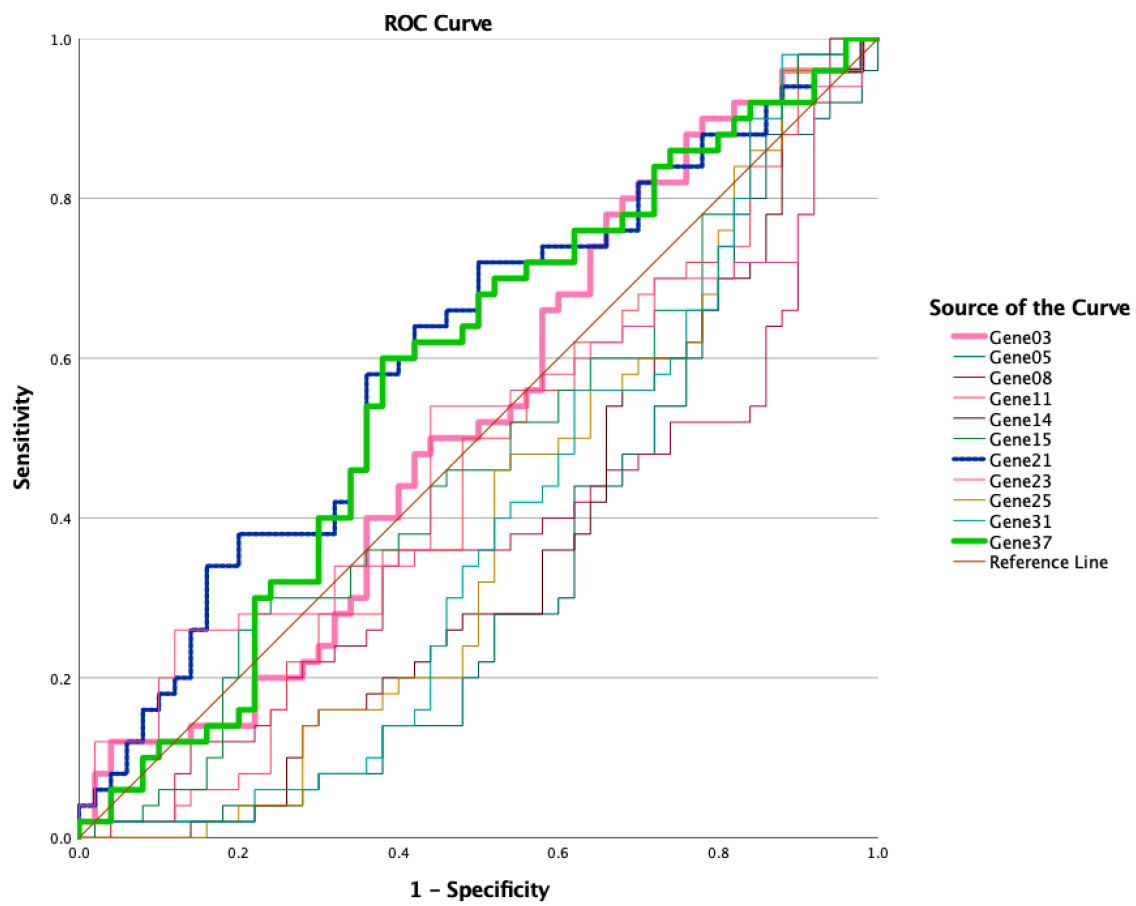


Table 39 Significant Predictors: Model 2

	B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I. for EXP(B)	
							Lower	Upper
Gene03	0.519	0.179	8.374	1	0.00	1.68	1.182	2.386
Gene37	0.616	0.185	11.111	1	0.00	1.852	1.289	2.66
Gene05	-0.469	0.178	6.913	1	0.01	0.626	0.441	0.887
Gene11	0.348	0.135	6.644	1	0.01	1.416	1.087	1.845
Gene25	-0.462	0.203	5.183	1	0.02	0.63	0.423	0.938
Gene14	-0.666	0.257	6.685	1	0.01	0.514	0.31	0.851

Table 40 Overall Prediction Accuracy: Model 2

		Predicted Cancer		Percentage Correct
		Healthy (0)	Non-Healthy (1)	
Observed Cancer	Healthy (0)	40	10	80
	Non-Healthy (1)	12	38	76
Overall Percentage				78